

Navigating the Rugged Data Landscape: The Impact of Data-Extrapolation Technologies on Knowledge Production

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March 25, 2026

Abstract

Data-extrapolation technologies enable progress by importing and applying empirical patterns from one domain to another. Their reliance on empirical patterns, however, is also their limitation: these tools can only be applied where data is abundant. Using the setting of structural biology, I investigate the impact of a data-extrapolation technology on innovation. I exploit the variation in prior data availability to compare areas where the tool was usable to those where it was not. I find that the tool increased the quantity of research in data-rich areas, driven primarily by new entrants: the resulting output yielded limited biological insights but was methodologically rigorous and contributed to downstream research.

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1. Introduction

Data-extrapolation technologies—tools that import and apply insights from past data—are becoming ubiquitous. By identifying empirical patterns from one context and extending them to another, these tools can enable progress and reduce the need for formulating theory-driven hypotheses or generating new experimental observations if similar data already exists.

Such tools range in sophistication. The simplest approaches rely on direct pattern matching with minimal adjustments.¹ More formal statistical methods, such as regressions, estimate relationships from past data by fitting pre-specified functional forms. Modern machine learning algorithms take this further by automatically discovering high-dimensional representations from training data that can be applied to previously unseen examples.²

While these data-extrapolation technologies translate empirical patterns across contexts with varying complexity, they share a critical constraint: they cannot function without a stock of prior data. Unlike simulation tools, which generate predictions based on theoretical principles and can be applied in any setting governed by those principles, data-extrapolation tools struggle to generalize beyond what past observations contain. Their ability to extrapolate also improves with greater data availability, in contrast to general computation tools that execute identical operations regardless of dataset size.

Because data-extrapolation tools depend on prior data, they enable innovation unevenly, unlocking progress in data-rich but not data-poor domains. In computer vision, for example, facial recognition has advanced rapidly due to the wealth of online images (Solon 2019), while medical imaging AI—despite its clinical importance—has faced steeper barriers to progress, in part because regulations limit data access (Koetzier et al. 2024). Large language models benefited from the vast corpora of digitized text and have emerged as the dominant form of AI today, whereas intelligent robotics—though once central to early visions of AI—has lagged, partly due to the dearth of video data on physical interactions (Gibney 2024). The trajectory of AI has been shaped as much by data availability as by the scientific potential of its applications.

¹ For example, before the rise of physics-based, numerical weather predictions, meteorologists relied on analog methods, identifying similar atmospheric patterns from historical weather maps to forecast the weather (Lynch 2008, 2018).

² Machine learning literature distinguishes between “interpolation” (prediction within the range of the training data) and “extrapolation” (prediction outside it). While machine learning algorithms are typically described as interpolating, some argue that they also exhibit forms of extrapolation (Balestriero, Pesenti, and LeCun 2021). In this paper, I use “data-extrapolation technologies” broadly to describe any tools that transfer empirical patterns, whether within or beyond the training distribution.

As data-extrapolation tools become central to innovation, understanding how they interact with the data landscape becomes essential. Although large-scale mapping efforts aim for comprehensive coverage (Williams 2013; Nagaraj 2022; Kao 2024), most data are unevenly generated, shaped by past priorities and constraints (Nagaraj and Stern 2020). By lowering barriers only in data-rich domains, data-extrapolation tools can steer innovation toward where data is abundant. The impact of such discoveries therefore depends on what problems remain in data-rich domains, which may have been empirically mapped due to their importance or tractability. Beyond redirecting effort, data-extrapolation technologies may also amplify the value of data itself: even datasets that do not initially provide any insights can serve as training data for future extrapolation, raising the returns to collecting data where immediate payoffs are limited.

Empirically studying the impact of data-extrapolation technologies on innovation, however, presents several challenges. To assess whether such a technology shifts the direction of innovation, one must observe not only realized research projects but also those that could have been pursued in the absence of the technology—an often unobservable counterfactual. There must also be a credible way to measure the availability of past data, as well as the similarity between potential projects, since extrapolation technologies work by applying insights from one past project to a new but related project. Finally, this technology must differentially treat only some areas of the setting, such that outcomes in the areas where the technology was introduced can be compared to the areas without the technology.

I focus on the setting of structural biology, a field with empirical features ideally suited for this paper. Structural biology studies the 3D structures of proteins and has contributed to more than a dozen Nobel prizes, as proteins play vital roles in nearly all biological processes. Elucidating a protein structure at atomic resolution—or “solving” the structure—can reveal the protein’s function, which helps with applications such as designing vaccines that target the spike proteins of SARS-CoV-2. Crucially, structural biology offers several empirical features that allow me to identify how a data-extrapolation technology may have shaped the rate and direction of innovation.

First, unlike many settings where only realized projects are observable, structural biology provides a window into the entire knowledge landscape. Using a database of all known proteins, I observe which proteins structural biologists chose to explore and which they left unexplored. In addition, similarity between projects can be quantified: proteins are composed of amino acid sequences, and they can be grouped based on their sequence similarity. This makes it possible to map out the landscape of all known proteins and see which areas of the landscape have been

explored (which I term “bright” clusters of proteins) and which areas remain unexplored and do not have an accumulation of prior structural data (which I term “dark” clusters).

Second, structural biology is well-suited for studying data-extrapolation technologies. Solving a protein structure involves extensive knowledge of biology, physics, and statistics, but many of the steps have now become automated. The specific technology I examine is the software program Phaser, released in 2003, which automates a method called molecular replacement (MR). Rather than solving a structure from scratch, MR borrows templates from previously solved proteins with similar amino acid sequences to help determine new structures.

Finally, this data-extrapolation technology differentially treated only parts of the knowledge landscape. Since MR relies on prior structure data, MR works in bright clusters of proteins (i.e., clusters with previously solved structures) but not in dark clusters. This enables a difference-in-differences design where bright and dark clusters serve as the treatment and control groups. By linking Swiss-Prot (a database of all known proteins) to the Protein Data Bank (a database of solved protein structures), I examine changes in the quantity and quality of solved structures in bright clusters after the arrival of MR, relative to dark clusters.

My first set of results focuses on the volume of innovation. MR reduced the cost of solving proteins in bright clusters, and this reduction in cost—perhaps unsurprisingly—increased the total number of structures solved in those clusters. This effect was sustained throughout the sample period: bright clusters turned increasingly brighter.

I then examine how MR impacted quality and distinguish between two dimensions of quality: importance (does this project address a problem of scientific significance?) and execution (is this project methodologically rigorous?) Both are dimensions on which the scientific community evaluates research, with strength in one dimension often helping offset limitations in the other. In structural biology, structures are typically not solved for their own sake but to generate new insights. Importance refers to whether the structure led to a novel understanding of a biological process, while execution refers to how meticulously a structure was elucidated (e.g., the resolution or the level of detail found in the structure).

In terms of importance, bright clusters disproportionately received structures that were scientifically less meaningful after MR. These structures yielded fewer functional annotations, were cited less frequently by patents, and had lower publication impact. Suggestive evidence leveraging variation in technical difficulty across bright clusters indicates that the topology of bright clusters—which proteins remained unsolved when MR arrived—shaped these results.

As for execution, however, I find that the number of well-executed structures increased. If MR primarily enables low-importance proteins, execution quality may become especially salient, as scientists place greater emphasis on technical rigor. MR also directly raises the value of execution: well-executed structures can serve as high-quality templates for future extrapolation, making technical rigor valuable independent of whether the structure itself provides a new biological understanding. Indeed, proteins in bright clusters were frequently mentioned in the text of downstream articles, even if not formally cited. This suggests that these structures may not have had broad citation impact but still served as useful technical inputs for follow-on research.

Finally, to assess the net impact of MR, it is also important to ask whether MR led to an overall expansion of research in structural biology or instead reallocated effort away from dark to bright clusters. Reallocation is a particularly relevant concern with MR: solving the first structure in a dark cluster generates data that make extrapolation possible. Reallocation away from dark clusters therefore forgoes not only immediate discoveries but also spillovers that enable extrapolation to the rest of the dark cluster.

Understanding who drove the increase in bright clusters helps distinguish between expansion and reallocation. If new entrants drove the increase, this suggests expansion through broadened participation. If the increase came from incumbents, the implications depend on how much MR shifted the relative attractiveness of bright over dark clusters. I find that the increase in bright clusters was largely driven by scientists from non-elite institutions and those with limited prior experience in structural biology, suggesting that MR lowered entry barriers and attracted new entrants rather than diverting effort away from dark clusters.

A potential identification concern is that bright and dark clusters may have been evolving on different trends before MR. First, while bright clusters may have been initially targeted because they were more important or more tractable, such differences in levels do not threaten the difference-in-differences design, as long as pre-trends are parallel, which I verify in the event study figures. Second, I control for predicted brightness based on ex-ante traits related to biological importance and technical feasibility. The goal is to compare clusters that are similar ex-ante but happened to differ in whether they were bright or dark when MR arrived. Even after controlling for predicted brightness, the effect of actual brightness remains significant.

Taken together, these results underscore both the benefits and the limitations of data-extrapolation technologies. These tools reduce the cost of research but only where prior data exists; the data landscape shapes which problems these tools can unlock. In the case of MR, bright clusters experienced a sustained increase in solved structures, primarily by new entrants, but these

structures yielded limited biological insight. At the same time, they were well-executed and frequently referenced downstream as technical inputs, suggesting that even incremental structures can generate value. More broadly, these discoveries contribute to the stock of data and may prove valuable once data-extrapolation tools are available to repurpose them as training templates.

This paper contributes to several lines of research. I first build on a body of evidence that examines how tools can both advance and constrain knowledge production. Studies have shown that when innovation becomes cheaper through new tools or research infrastructure, this lowers barriers to entry and sparks diverse follow-on research (Furman and Stern 2011; Teodoridis 2018; Nagaraj, Shears, and De Vaan 2020). At the same time, other studies caution that reliance on standardized tools can lead to less novelty or opportunities for differentiation (Mannucci 2017; Miric, Ozalp, and Yilmaz 2023; Impink and Wright 2025). The question is whether the tool expands the set of problems that can be tackled or primarily enables exploitation of existing problem space (Barbosu and Teodoridis 2024).

I contribute to this literature on research tools by focusing on the role of data availability in data-extrapolation technologies (Agrawal, Gans, and Goldfarb 2018; Q. Hao et al. 2026). Existing work on the limits of such tools has primarily focused on two factors: the capabilities of the algorithm itself—how far it can extrapolate from what is known and the accuracy of its predictions—and the quality of the training data, where flawed data can lead to bias (Cowgill et al. 2020; Choudhury, Starr, and Agarwal 2020). This paper highlights a related but distinct factor: the uneven data landscape. Independent of algorithmic capability or data quality, variation in data availability dictates where data-extrapolation technologies can be applied at all. Only some problems can benefit from algorithmic discovery while others are excluded altogether due to insufficient data (Cockburn, Henderson, and Stern 2018).

I also build on the literature on knowledge geography (Kauffman 1993; Stuart and Podolny 1996; Levinthal 1997; Fleming and Sorenson 2004). Recent theoretical work includes Carnehl and Schneider (2025), who model scientists’ choice between expanding the frontier or deepening knowledge; building on this framework, Gans (2025) investigates how AI may alter this tradeoff. A particularly relevant contribution is Hoelzemann et al. (2024), who study distortions in data generation and illustrate the “streetlight effect” of data, in which data on moderately promising options can deter exploration.

This paper complements the prior literature by providing an empirical measurement of the knowledge landscape and studying the impact of a data-extrapolation tool. I leverage the setting of structural biology, which was first brought to the attention of social scientists by Hill and Stein

(2025b, 2025a), and has since emerged as an attractive setting for innovation research given the field’s rich scientific achievements and unusually detailed data (e.g., Zhuo 2023). A key empirical contribution of this paper is the construction of a map of the knowledge landscape, which is challenging to observe in other settings.

The rest of the paper proceeds as follows. Section 2 provides a conceptual framework on the impact of data-extrapolation technologies on innovation. Section 3 introduces the context of structural biology, and Section 4 describes the main data sources. Section 5 outlines the difference-in-differences design that underpins this study’s empirical strategy. Section 6 presents the results, while Section 7 discusses the broader implications and concludes.

2. Conceptual Framework

This section examines how data-extrapolation technologies interact with the data landscape. I first identify two key features of the data landscape that data-extrapolation tools amplify. I then outline a simple framework for understanding how these tools may shape the importance of the innovations that get unlocked, the rigor with which they are executed, and who produces them.

2.1 Data-Extrapolation Technologies and the Data Landscape

2.1.1 Data Landscape

Data is the product of past priorities and constraints. Governments deploy censuses and surveys to measure demographic and economic trends, subject to budgetary restrictions. Scientists collect observations and conduct experiments, driven by research agendas and methodological feasibility. Firms track customer behavior to improve sales within the limits of privacy regulations and technical infrastructure. All datasets—even those that aim for comprehensive coverage—are curated, balancing the value of the information gained against the cost of collecting the data (Gitelman 2013; Wiggins and Jones 2023). Data is therefore unevenly generated, reflecting not only what is important but also tractable.

A second feature of the data landscape is that data can generate value, even when its original purpose has been exhausted or if it does not offer immediate insights. Datasets are often repurposed, from administrative census files serving as inputs into economics research (Nagaraj and Tranchero 2024) to electronic health records collected for clinical care being used in scientific research (Wang 2025). This is one of the key arguments for “Big Data” projects—large-scale data

collection efforts that may not be driven by clearly defined hypotheses but could prove valuable in the future. The Human Genome Project is one such example that catalyzed many advancements in biology, though early detractors once criticized the project as “mediocre science” with a “brute force” approach that will yield “no interesting biological information” (Utz 2024).

2.1.2 Data-Extrapolation Technologies

Data-extrapolation technologies import and apply empirical patterns from one domain to another, uncovering connections across disparate areas (Agrawal, Gans, and Goldfarb 2018; Agrawal, McHale, and Oettl 2018) and detecting correlations without requiring full causal understanding (Mullainathan and Rambachan 2024; Tranchero 2023a, 2023b). Data-extrapolation technologies offer a form of shortcut: they can lower the cost of research by reducing the need for theory-driven hypotheses or new experimental observations if prior, similar data already exists. In drug discovery, for instance, machine learning models can help identify promising molecules, without requiring a mechanism of action or experimental screening (Stokes et al. 2020).

Yet this reliance on empirical patterns is also their limitation: unlike simulation tools based on theoretical principles or general computation tools whose performance is invariant to dataset size, data-extrapolation technologies can only perform well if there is sufficient data of prior templates. This “unreasonable effectiveness of data” (Halevy, Norvig, and Pereira 2009) means that the ability of data-extrapolation technologies depends on the data landscape, beyond its algorithmic capability.

Specifically, data-extrapolation technologies interact with the two features of the data landscape. First, because data is unevenly generated, the data landscape shapes which problems become viable for data-extrapolation tools. While some datasets are deliberately built to unlock these tools’ potential, such efforts are also subject to tractability barriers that limit where data can be produced.³ As a result, the impact of these tools depends on where data accumulates or can feasibly accumulate. Second, data-extrapolation technologies increase the potential for data to generate value beyond its original purpose; they expand the use of existing data, such as online images and digitized books that now underpin computer vision and language models. Above all, data-extrapolation tools reshape what can qualify as valuable data: datasets that seem to offer

³ ImageNet, a large-scale database of annotated images, is perhaps the most influential dataset that was explicitly constructed to stimulate AI research in computer vision. However, it was the existing availability of online images that made ImageNet possible in the first place (Denton et al. 2021).

few scientific insights can later prove critical once data-extrapolation tools are available to repurpose them.

Taken together, this paper focuses on two channels that link data-extrapolation tools to the data landscape: (i) these tools lower the cost of research by importing empirical patterns but can only work where data accumulates, and (ii) they can alter incentives for the collection of new data as they amplify the repurposing potential of data. These two channels motivate the following framework on how data-extrapolation technologies may impact innovation.

2.2 Data-Extrapolation Technologies and Innovation

2.2.1 Importance

By reducing research costs in data-rich domains, data-extrapolation technologies increase the volume of viable projects and can drive innovation where data is abundant. Under what conditions will this facilitate discoveries that yield new scientific insights? While their underlying algorithmic capability determines how far the tools can extrapolate from existing templates to new cases (Chollet 2019; Balestriero, Pesenti, and LeCun 2021; Dell’Acqua et al. 2023), the topology of the data landscape also determines whether such opportunities exist in the first place.

If data-rich domains contain valuable unsolved problems, data-extrapolation tools may enable important discoveries. This could occur if these areas were prioritized for data collection due to their research potential, but high costs prevented some projects from being pursued. In other cases, data-rich domains may contain “hidden gems”—valuable problems that are neglected, for example, due to priority incentives that steer researchers toward novel areas rather than empirically mapped territories (Merton 1957), or because of imperfect ability to predict which problems will prove fruitful (Aguiar and Waldfogel 2018). By lowering costs, data-extrapolation technologies can make these projects attractive.

On the other hand, these tools may not lead to breakthroughs if data-rich areas contain few important projects. This could arise if prior data accumulation was driven primarily by tractability.⁴ Moreover, since data-rich domains are areas that have been at least partially

⁴ The idea that scientists consider both the tractability and importance of a project is not new (Stephan 2012). Examples of tractability-driven data can be found across many fields: benchmark datasets in reinforcement learning that rely on environments such as Atari video games that generate data cheaply but may not translate well to real-world problems (K. Hao 2018); the proliferation of surrogate endpoints in clinical research that are easier to measure and collect, instead of outcomes such as quality of life (Budish, Roin, and Williams 2015; Wallach et al. 2024); and the “college sophomore problem” in social psychology,

explored, it is possible that the most consequential lines of inquiry have already been exhausted (Bloom et al. 2020). In such cases, additional innovations enabled by data-extrapolation technologies may have limited impact.

While examining how past priorities and constraints shape the evolution of the data landscape is beyond the scope of this paper, the key insight is that uneven data availability, distinct from algorithmic capability, can moderate the impact of data-extrapolation tools on innovation.

2.2.2 Execution

Beyond importance, data-extrapolation technologies can also affect execution quality—the rigor and precision with which a given problem is solved. Importance and execution are distinct dimensions of research quality: a project can address an important problem with weak methods, or be methodologically rigorous but solve a trivial problem.

By shaping the importance of the problems that can be tackled in the uneven data landscape, data-extrapolation tools can alter the emphasis placed on methodological rigor. Successful publications or grant applications typically require merit along both dimensions of importance and execution, with strength in one dimension helping offset weakness in the other.⁵ If data-extrapolation technologies primarily unlock problems of lesser importance, the relative value of rigor rises, as journals and funders assign greater weight to methodological quality. In contrast, if these technologies enable access to important problems, researchers can compete on generating new discoveries, reducing the relative emphasis on execution quality.⁶

In addition, data-extrapolation technologies can impact execution quality more directly due to the repurposing potential of data. Prior to these tools, technically rigorous work that did not produce novel insights was valuable to the extent it established reliability for existing findings. Data-extrapolation technologies, however, introduce an additional incentive for producing rigorous

where a large share of data relies on undergraduates who are easy to recruit as lab participants for course credit (Sears 1986; Cooper, McCord, and Socha 2010).

⁵ For example, the NIH explicitly asks reviewers to score grant applications on the criteria of “Importance” and “Rigor and Feasibility,” in addition to an “Overall Impact” score. NIH’s guidance on assessing the overall impact discusses the balance between importance and execution: “[I]f the potential significance of a study was so great as to overshadow a number of methodological weaknesses then this should be clearly stated. Likewise, if the design of the study is so flawed as to negate any potential significance and/or innovation of the study then this should be clearly stated” (NIH Center for Scientific Review 2018).

⁶ This discussion implicitly assumes that importance is uncorrelated with intrinsic problem difficulty. If more important problems are harder to execute well, the relationship between importance and execution becomes more complex.

work: projects that fail to produce new insights can still generate data that become inputs for extrapolation. If researchers internalize this spillover, data-extrapolation technologies may directly incentivize greater rigor by raising the downstream value of high-quality data.

2.2.3 Who Innovates?

Reduction in research costs impacts participation on two margins: the entry of new contributors and changes in effort allocation by established researchers in the field (Furman and Teodoridis 2020; Nagaraj et al. 2020). For data-extrapolation tools, the uneven distribution of data determines where entry becomes possible for new contributors, while shifting the attractiveness of data-rich domains relative to data-poor ones for incumbents. Moreover, the repurposing potential of data can increase the cost of neglecting data-poor domains.

New researchers typically face higher costs than incumbents due to limited expertise and resources, which can make participation prohibitively expensive. However, when data-extrapolation tools reduce costs, entry into data-rich regions becomes viable, though data-poor regions remain inaccessible. For researchers excluded from the field, even projects of limited importance in data-rich regions may attract entry once costs fall enough to yield a positive return.

For incumbents already in the field, their response depends on the relative attractiveness of data-rich and data-poor domains. Cost reductions by data-extrapolation technologies may be insufficient to substantially shift efforts towards data-rich domains if those areas primarily contain low-importance problems. Conversely, when these domains contain high-importance problems, incumbents may expand efforts, at the expense of reducing activities in data-poor domains if capacity cannot be scaled.

This reallocation by incumbents is not necessarily problematic if expected returns in data-rich areas exceed those in data-poor areas. However, if information about project returns is imperfect, reduced exploration of data-poor areas may lead incumbents to overlook unanticipated, high-value opportunities. Moreover, recent work examines how data spillovers may distort exploration incentives (Hoelzemann et al. 2024). Data-extrapolation technologies can compound the spillover effects of data: these tools increase the value of generating data in data-poor areas since solving an initial problem creates data that can be extrapolated to related problems, unlocking previously inaccessible regions of the problem space. Reallocation away from data-poor regions thus may come at the cost of losing both direct discoveries and spillovers that enable future work.

2.2.4 Summary

Data-extrapolation technologies can shape the quality of innovation along two dimensions—the importance of the problems that are pursued and the rigor with which they are executed—as well as who produces them. The tools’ interaction with the uneven data landscape plays a central role by determining which problems become viable and by extending the repurposing potential of data. How these tools shape innovation is ultimately an empirical question, and the goal of this paper is to provide evidence on these dynamics.

3. Empirical Setting

An ideal empirical setting needs three ingredients: (i) an observable knowledge landscape, where one can track which research projects get explored versus unexplored, as well as a measure of similarity between potential projects, (ii) the arrival of a data-extrapolation technology, and (iii) specifically, the differential arrival of the technology, such that it only arrives in some parts (treated) but not other parts of the setting (control). In this section, I introduce the setting of structural biology and discuss the field’s scientific importance as well as empirical features that are well-suited for this paper.

3.1 Structural Biology: The Study of Proteins

Structural biology is a field that studies the 3D structures of proteins and aims to uncover the functional roles of proteins by elucidating their structures. As Francis Crick (who discovered the helical structure of DNA) remarked, “If you want to understand function, study structure” (Crick 1990). Since proteins are responsible for carrying out most functions in cells, insights from structural biology have helped with a broad range of applications, from identifying targets for new drugs to understanding disease progression. As one indication of its wide-reaching impact, structural biology has been recognized with more than a dozen Nobel prizes.

Recently, structural biology played an important role in the efforts against the coronavirus pandemic. As shown in Appendix Figure 1A, researchers solved the structure of the spike proteins on the surface of SARS-CoV-2, determining the 3D coordinates of the protein’s individual atoms. Through this direct visualization, the scientific community learned how these proteins latch onto receptors on human cells like “a key to a lock” (Patel, Lucet, and Roy 2020), enabling the development of vaccines that are designed to block these proteins.

3.2 Structural Biology as an Empirical Setting

3.2.1 Observable Knowledge Landscape

In order to investigate whether a data-extrapolation technology changes the direction of innovation, it is important to observe the entire knowledge landscape. In most settings, however, researchers can only observe projects that were realized, while alternative projects that could have been pursued (but were not) remain invisible. An attractive feature of structural biology is that it provides a unique window into this landscape. As detailed in Section 4, I leverage a database of all known proteins, and I can observe which proteins structural biologists chose to explore versus could have explored but neglected.

Furthermore, quantifying the similarity between each potential project is challenging. For instance, in the case of scientific publications, the distance between each paper is difficult to assess; text similarity is often used, but this is an imperfect proxy for measuring intellectual distance. In contrast, structural biology provides an objective measure: proteins are composed of sequences of organic compounds called amino acids—which are given by nature—and proteins can be grouped based on their sequence similarity.⁷ Since data-extrapolation technologies work by identifying similarities from one project to another, this measure of distance enables me to track the use of such extrapolation.

3.2.2 Arrival of a Data-Extrapolation Technology

Structural biologists developed various experimental techniques to reveal the atomic structure of proteins. Solving a protein structure involves deep knowledge of biology, physics, and statistics, and this used to be—and remains—challenging. A complex structure could take months, even years, to solve. For instance, determining the structure of the ribosome (a macromolecular machine responsible for translating DNA code to produce proteins) took over two decades, culminating in the 2009 Nobel Prize in Chemistry (Ramakrishnan 2018).

⁷ I build on the works of Hill and Stein (2025b, 2025a), who study the effect of competition in structural biology and cluster structures based on their sequence similarity to identify scientists engaged in “priority races” (i.e., competing teams that worked on structures in the same cluster, unbeknownst to each other). In this paper, rather than focusing on just proteins whose structures have been characterized, I look at the entire universe of proteins—both structurally characterized and uncharacterized—and cluster this universe of proteins based on their sequence similarity.

The dominant method of solving a structure is X-ray crystallography,⁸ which proceeds in three main steps (Appendix Figure 1B). First, the protein sample must be crystallized, arranged into a well-ordered crystal lattice, consisting of multiple copies of the protein. Second, once the crystal is obtained, X-ray beams are directed at the crystal, which produces diffraction patterns, as electrons in the crystal scatter the X-ray. Third, using a combination of physical laws, statistics, and intuition, structural biologists construct a density map of electrons from the diffraction patterns and build a 3D atomic model of the protein structure. I focus on this third step of interpreting the diffraction data. Since the time of Max Perutz (co-winner of the 1962 Nobel Prize in Chemistry), who solved the first protein structure through painstaking hand-calculations, many steps of interpreting the diffraction data have become automated.

The specific technology I examine is the software program Phaser. Phaser was released in September 2003, which automates a method called molecular replacement (MR). Figure 1 shows the rise in the number of structures solved by MR at the Protein Data Bank, a global repository of all structures.⁹ One of the central challenges in interpreting the diffraction data is the “phase problem,” a problem sufficiently difficult that one method for addressing it resulted in a Nobel prize.¹⁰ Prior to MR, structural biologists had to resort to costly experimental methods to overcome the phase problem,¹¹ but MR allowed structural biologists to bypass experimental phasing. Instead of determining the phases from scratch, MR relies on a previously solved structure that shares close sequence similarity with the unknown structure. This prior structure serves as a template to recover the phases of the unknown structure.

MR is therefore a data-extrapolation method: it imports phase information from neighboring proteins, exploiting the empirical pattern that sequence similarity predicts structural similarity,

⁸ In addition to X-ray crystallography, two other methods can be used to solve a structure: nuclear magnetic resonance spectroscopy and cryo-EM. However, crystallography is by far the most common method, as 90% of protein structures are solved using this method during my sample period.

⁹ The method of MR was first proposed in 1962, but MR was not put into wide practice until decades later due to lack of available structures, as well as lack of ready-made software programs (Doerr 2014). While Phaser was not the first software program to implement MR, it is the most widely-used program, as it is user-friendly and considered to be the most efficient (Scapin 2013).

¹⁰ X-ray reflections have both amplitudes and phases, but the phase cannot be inferred from the diffraction patterns. Without knowing the phase, a model of the protein structure cannot be constructed. Max Perutz and John Kendrew were awarded the 1962 Nobel Prize in Chemistry, in part for their pioneering work in overcoming this phase problem.

¹¹ There are two experimental methods that solve the phase problem, which do not rely on the availability of prior solved structures. See Appendix B for more details.

although the causal mechanisms through which amino acid sequences determine a protein’s 3D structure (a process called “protein folding”) are not fully understood.^{12,13}

3.2.3 Differential Arrival of a Data-Extrapolation Technology

Finally, MR arrived in some parts of structural biology but not others. Since I observe the map of known proteins and the distance between each protein in terms of their sequence similarity, I can identify which clusters of proteins received attention from structural biologists before MR (“bright” clusters of proteins) and which did not (“dark” clusters). Because MR requires data on previously solved structures, it can be applied in bright clusters but is not useful for dark clusters. Bright and dark clusters therefore serve as the treatment and control groups, respectively, providing the basis for a difference-in-differences design described in Section 5.

4. Data

To map the landscape of potential proteins that structural biologists can target, I rely on two main datasets: UniProt/Swiss-Prot, a database of all known proteins, and the Protein Data Bank, a database of all publicly available protein structures. I then cluster proteins based on their sequence similarity to construct the final sample.

4.1 UniProt Knowledgebase/Swiss-Prot

The Universal Protein Resource Knowledgebase (UniProt) is a comprehensive database of proteins. Information for making a protein is stored in a gene’s DNA, and by translating the DNA sequence of a gene, scientists can determine the protein’s existence and the sequence of amino acids that appear in the protein. Protein sequences in UniProt are thus sourced by translating genes from major genome sequence databases.

¹² While MR and machine learning methods are both data-extrapolation approaches that rely on the existence of prior data from which they import information, the principles behind MR and machine learning are distinct. Appendix B.3 provides details.

¹³ In November 2020, a technology that supersedes MR was introduced: the AI program AlphaFold, created by Google’s DeepMind. AlphaFold can predict the structure of a protein based purely on its sequence of amino acids. While MR helps with specifically the phase problem of experimental structure solving, AlphaFold bypasses the need to conduct experiments at all. While AlphaFold’s success falls outside of the time period studied in this paper, I discuss potential implications in Section 7.

To define the complete set of proteins at risk of being structurally characterized, I focus on the Swiss-Prot section of UniProt.¹⁴ Created in 1986, the Swiss-Prot database is extensively reviewed, maintained, and annotated by experts based on experimental results and literature review. As of October 2020, Swiss-Prot contains 563,552 protein entries.

4.2 Protein Data Bank

Established in 1971, the Protein Data Bank (PDB) is a repository of protein structures and contains approximately 170,000 structures as of October 2020. Since the early 1990s, most journals have required authors to deposit their structures in the PDB as a requirement for publication (Berman et al. 2000); the PDB therefore contains the universe of all publicly available structures. The PDB provides detailed descriptions of each structure, as well as crosswalks to Swiss-Prot.

4.3 Sample Construction: Clustering Proteins

After identifying which proteins in Swiss-Prot were found to be structurally characterized in the PDB, the final step is to measure the distance between each protein and cluster proteins that share sequence similarity. I rely on MMseqs2,¹⁵ an algorithm used by both Swiss-Prot and the PDB to cluster similar proteins (Steinegger and Söding 2018; Hauser, Steinegger, and Söding 2016). Given that MR will likely be successful if the template and the target proteins share at least 30% sequence identity (Schmidberger et al. 2010; Phenix 2022), I chose a threshold of 30% sequence identity to group all proteins in Swiss-Prot into mutually exclusive clusters. I then restricted the sample to clusters with at least one human protein and clusters that had at least one protein discovered by 1998, the year before the panel begins. More details on sample construction can be found in Appendix A.

¹⁴ In addition to Swiss-Prot, UniProt has a database called TrEMBL, which is larger but contains computationally annotated proteins whose existence is largely not experimentally proven. More details are provided in Appendix A.1.

¹⁵ MMseqs2 can be downloaded from <https://github.com/soedinglab/MMseqs2>. More details on MMseqs2 are provided in Appendix A.3.

5. Empirical Strategy

5.1 Main Specification

Since MR relies on having similar, previously solved structures as templates, MR can only be applied in clusters of proteins with previously solved structures (i.e., bright clusters) and does not work for clusters of proteins that have not yet been structurally characterized (i.e., dark clusters). This enables a difference-in-differences approach, where I estimate the following regression equation to examine the impact of MR:

$$Y_{ct} = \beta_0 + \beta_1 PostMR_t \times Bright_c + \delta_t + \gamma_c + \varepsilon_{ct} \quad (1)$$

Y_{ct} is the total number of structures that gets solved in cluster c in year t . $PostMR_t$ is an indicator variable that turns one after the arrival of MR in 2003, and $Bright_c$ is an indicator variable for bright clusters, defined as whether the cluster had at least one structure by 1998.¹⁶ δ_t are calendar year fixed effects, and γ_c are cluster fixed effects. β_1 is the coefficient of interest and can be interpreted as the impact of MR on the number of solved structures. Standard errors are clustered at the cluster level.

Identification relies on the parallel trends assumption: in the absence of MR, outcomes for bright and dark clusters would have followed similar trends, conditional on cluster and year fixed effects, as well as additional time-varying controls included in some specifications.

5.2 Descriptive Statistics

As shown in Table 1, my sample consists of 6,942 clusters, with 9% of the clusters classified as bright. The table summarizes several features related to the technical feasibility and biological importance of the proteins in the clusters at the time MR was introduced. The difficulty of solving a protein is determined by the protein’s physicochemical properties, including whether the protein

¹⁶ The treatment variable, $Bright_c$, is defined as whether the cluster had a structure by 1998 (the year before the panel begins) instead of 2003 (when MR arrived). If $Bright_c$ is defined using the year 2003, then the treatment is mechanically correlated with the outcome variable (the number of structures being solved each year) in the pre-period from 1999 to 2003 since the treatment is a lagged outcome of the pre-period. The panel was chosen to begin in 1999 because this is (i) early enough to yield at least five years of pre-period before the introduction of MR, but (ii) late enough that there has been some accumulation of prior solved structures in the PDB (6% of structures that will eventually be deposited at the PDB by 2019 had accumulated by 1998).

has membrane regions, intrinsic disorder, and compositional bias, as well as its sequence length.¹⁷ To assess biological importance, I use several proxies: the number of publications and drugs associated with the protein, whether the protein has a known biological function or disease relevance, and whether it is of human origin.

In terms of levels, bright clusters were composed of proteins that were, on average, both easier to solve and more biologically important. For example, the average bright cluster had a lower proportion of disordered and membrane proteins—both known to hinder structure determination—compared to dark clusters. Bright clusters also exhibited a higher share of proteins with known biological function, as well as a greater number of associated publications and drugs.

These descriptive patterns are consistent with bright clusters reflecting areas that scientists found both important and tractable. While these differences in levels between bright and dark clusters do not threaten the difference-in-differences strategy—as long as pre-trends are parallel—I revisit these characteristics in Appendix A.6, where I construct and control for a “predicted brightness” measure based on pre-period traits related to both importance and feasibility.

6. Results

6.1 Impact of MR on the Quantity of Innovation

I begin by examining how MR impacted the number of solved structures. As shown in Table 2, bright clusters turned brighter (i.e., received more structures) after MR, relative to dark clusters. The outcome is the total number of solved structures in a cluster each year. Columns 1-2 report the outcome after Log(+1) transformation,¹⁸ while Columns 3-4 report the results in levels (scaled by the standard deviation). As reported in Column 1, bright clusters experienced a 7% increase in the number of solved structures after the arrival of MR, relative to dark clusters. Results in levels also indicate that bright clusters turned brighter. These clusters received an increase of 0.73

¹⁷ Membrane proteins, which are embedded in or associated with cell membranes, tend to be flexible and partially hydrophobic—features that make them especially challenging to characterize structurally. Proteins with intrinsically disordered regions, which lack a stable three-dimensional conformation, are difficult to solve (Slabinski et al. 2007). Compositional bias—regions of the protein with overrepresented subsets of amino acids—is also linked to increased difficulty (Harrison 2017). Data on these features is primarily drawn from Perdigão et al. (2015), with additional updates based on my own data collection. Further details are provided in Appendix A.5.

¹⁸ In Appendix Table 1, I provide a robustness analysis using inverse hyperbolic sine transformation. Results remain similar.

annual number of structures after the arrival of MR, which translates to a 29.8% increase relative to the baseline standard deviation of 2.45.

I conduct several analyses to ensure that these results are driven by MR. First, one concern is that bright clusters may be getting more structures not necessarily due to MR but because they are also getting increasingly larger (i.e., more protein sequences are being discovered) or older. In Columns 2 and 4, I additionally control for time-varying cluster size and cluster age while estimating Equation 1.

Second, the impact of MR should be limited to structures that were actually solved using the method. If bright clusters exhibit increases in both MR-solved and non-MR-solved structures, this raises concerns that factors unrelated to MR may be causing bright clusters to turn brighter. In Appendix Table 2, I confirm that MR only increased the number of MR-solved structures.

Third, since the marginal benefit of an additional prior structure is likely greatest when going from zero to one, the impact of MR should be stronger when comparing dark clusters versus bright clusters with a single previously solved structure, and weaker when comparing bright clusters with a single versus multiple structures. Appendix Table 3 shows this result. I split the bright clusters into whether they had a single or multiple previously solved structures. I then examine the impact of MR, comparing dark versus bright clusters with just a single structure (Column 2) and comparing bright clusters with a single structure versus multiple structures (Column 3). The impact of MR is stronger in Column 2 relative to Column 3.

Fourth, to assess pre-period trends, I show an event studies version of Equation 1, replacing the single $PostMR_t$ indicator with indicators for every year before and after the introduction of MR. Figure 2 plots the dynamic effects of MR on the number of solved structures. In both Panels A (Log(+1) transformation) and B (levels), there appears to be no difference in pre-trends between bright and dark clusters. Moreover, the impact of MR is sustained over the sample period: bright clusters turned increasingly brighter.

In the Appendix, I provide additional robustness analyses. While the event studies show no evidence of pre-trends, one might still be concerned that bright and dark clusters followed different trajectories for reasons unrelated to MR. To address this, in Appendix A.6, I construct a predicted brightness measure to directly control for ex-ante traits. Appendix A.7 also presents alternative specifications, where I vary the cluster construction.

6.2 Impact of MR on Functional Insight

MR decreased the cost of solving structures in bright clusters, and, perhaps unsurprisingly, increased the volume of structures in those areas. The key question is whether this increase in structures led to a novel biological understanding. Structural biologists do not solve structures simply for the sake of solving them; they elucidate structures with the hope of learning a new functional insight. The editors of *Nature Structural Biology* advocated in their inaugural issue, “[T]he static image of the molecule is rarely an end in itself, but rather a beginning of comprehension” (Nature Structural Biology 1994). Through additional biochemistry or cell biology experiments, structural biologists link a protein’s structure to its potential function to understand the role the protein plays in various biological processes (Cassiday 2014).

To evaluate whether MR contributed new biological insights, I explore three outcomes. First, did the structure lead to a publication? Structures deposited in the PDB without an accompanying publication to explain their significance likely contributed only modestly to advancing biological understanding. As a prominent researcher at Yale once declared, “The fact is that protein structures come alive intellectually only when they are connected with [other data] indicating what they do” (Moore 2007).¹⁹ Second, was the structure cited by Swiss-Prot? The Swiss-Prot database contains extensive annotations about a protein’s function and provides references behind each annotation. Importantly, the references are added manually by experts who follow well-defined curation protocols, undergo quality checks, and are updated as new data becomes available. I track whether the publication associated with the structure was included as a reference under Swiss-Prot’s functional annotation section. Finally, I assess whether the structure was cited by a patent, as a proxy for having generated sufficient functional insight to enable downstream commercial applications. I leverage data from Marx and Fuegi (2020) on patent citations to scientific articles to identify structures with papers that were cited by at least one patent within five years of publication.

I find that bright clusters disproportionately received more structures that did *not* reveal functional insights after MR. As shown in Table 3, bright clusters received 8% more *unpublished* structures, which have no accompanying articles that describe their function (Column 1). In contrast, bright clusters experienced a marginal decrease in the number of structures that were

¹⁹ There could be several reasons why some structures do not have accompanying publications. A “stamp collection” of structures may be required to win grants from funding agencies. Some of these structures are also from structural genomics consortiums, whose goals are to catalogue as many types of structures as possible without necessarily explicating them (Petsko 2007).

cited by a patent (Column 4), and there were no differences between bright and dark clusters in the number of structures that were cited by Swiss-Prot (Column 6)—which are the set of structures that are the most likely to have yielded functional insights.

Complementing these analyses, I explored how the scientific community valued MR-enabled structures by analyzing their publication impact. Specifically, for each structure’s publication, I tracked their citation impact (Table 5) and journal prestige (Appendix Table 5),²⁰ and decomposed the number of solved structures into terciles based on each measure. I find that bright clusters especially received more structures with *less* publication impact. These patterns suggest that while MR increased structure output, it did not lead to work that the broader community deemed especially impactful.

One explanation is that the topology of bright clusters—which proteins remained unsolved when MR was introduced—restricted what was available. While examining the origin of bright and dark clusters before MR is beyond the scope of this paper, the descriptive statistics in Table 1 suggest that both tractability and importance played a role in the formation of bright clusters. This implies that bright clusters contain a mix: some may have been mapped primarily for tractability, leaving behind proteins of limited importance, while others may reflect areas of high potential where prohibitive costs prevented some proteins from being pursued.

To investigate this mechanism, I classify clusters based on their technical difficulty using a composite difficulty score constructed from the physicochemical protein features listed in Table 1.²¹ I then estimate the regression separately for easy and hard clusters to examine if MR’s effects vary by cluster difficulty. In easy clusters, high-importance targets are more likely to be depleted since cost is less of a constraint, while in hard clusters, important proteins may have remained unsolved due to high cost, which MR can make viable. Consistent with this pattern, Appendix Table 4 shows that in easy clusters, the increase in structures after MR was driven mainly by proteins that did not lead to published findings. In contrast, in hard clusters, MR unlocked proteins that were at least publishable, although these structures still offered limited value in terms of functional annotations or patent citations. While modest, these differences across cluster

²⁰ I linked each structure’s primary publication in the PDB to PubMed records and obtained citation data from OpenAlex (measured as the mean annual citations received within the first five years after publication), along with the journal’s impact factor.

²¹ I construct composite difficulty scores that range from 0 to 1 for each protein by averaging over the relevant features listed in Table 1 and aggregate these protein-level scores to the cluster level by calculating the mean. This approach has some caveats, as importance and difficulty may vary within a cluster and the composite scores have measurement limitations. See Appendix A.5 for details.

difficulty provide suggestive evidence that the composition of bright clusters may have played a role in determining what MR could uncover.

6.3 Impact of MR on Execution Quality

While MR appears to have primarily enabled the solving of proteins in bright clusters that did not generate novel biological insights, the resulting increase in incremental structures can still be valuable. In particular, a second dimension of quality to consider is execution—the rigor with which a scientific contribution is executed, independent of the significance of the problem it addresses. Especially when proteins do not reveal functional insights, structural biologists may place additional weight on producing structures that are methodologically rigorous to strengthen the scientific contribution of their work. Furthermore, the introduction of MR may incentivize scientists to produce high-quality structures, as they can serve as templates for future MR (Scapin 2013).

To measure execution quality, I rely on two metrics used by the structural biology community: resolution and R-free (Kleywegt and Jones 1997). Resolution captures precision, or the level of structural detail. Appendix Figure 1C shows an example of a protein (tyrosine 103 from myoglobin) at different resolutions, illustrating whether individual atoms can be distinguished. Resolution depends on the degree of order in the protein crystals, which researchers can influence by optimizing crystallization conditions. Highly ordered crystals—where proteins are tightly packed and uniformly aligned—produce diffraction patterns with finer details. The R-free refers to accuracy or goodness-of-fit: how well the model of the protein structure matches the observed experimental data. Structural biologists build atomic models of their protein structure, simulate diffraction patterns based on the models, and compare the simulated diffractions to the experimentally observed patterns. The R-free improves as researchers refine their models to better align with the experimental data.

In Table 4, I investigate the impact of MR on execution. Columns 1-3 decompose the total number of structures solved in a cluster into terciles based on the structure’s resolution, while Columns 4-6 similarly report results using the R-free.²² A clear pattern emerges: bright clusters especially received well-executed structures. For resolution, there was no difference between bright and dark clusters in the number of structures that were solved in the bottom tercile. In contrast,

²² For both resolution and R-free, lower values indicate higher quality. I construct terciles such that higher terciles correspond to higher quality.

bright clusters received 9% more structures that were solved in the top tercile, relative to dark clusters. Likewise, for R-free, bright clusters received just 4% more structures from the bottom tercile, but 9% more structures from the top tercile.

I then examine whether these structures in bright clusters provide value as technical inputs into downstream research, even in the absence of strong biological insights. I use a unique measure provided by the PDB called “mentions,” which tracks whether a structure is referenced in the text of downstream articles, even if not formally cited.²³ Mentions likely capture technical use cases, such as when structures are incorporated into methods, figures, or analyses. As shown in Table 5, MR did not significantly increase the number of highly cited structures, but it did disproportionately raise the number of structures that were frequently mentioned in downstream research. This pattern suggests that MR helped increase the supply of structures that, though not broadly influential in terms of citations, were still actively referenced and used by downstream researchers as technical inputs. Similarly, Appendix Table 5 shows that MR especially increased the number of structures published in lower impact journals and specialist structural biology journals, further suggesting that these contributions were technically valuable within the structural biology community, even if they were not recognized as high-impact by broader audiences.

6.4 Who Drove the Increase in Bright Clusters?

Whether entrants or incumbents drove the increase in bright clusters has implications for understanding how MR shaped participation in structural biology: did MR democratize the field by lowering technical barriers for entrants or did it primarily result in incumbents changing their effort allocation across bright and dark clusters?

Investigating who contributed to the growth in bright clusters requires identifying the principal investigator (PI) responsible for each structure. The PDB, however, reports only the authors’ last names and the initial of first names, with no author identifiers. I therefore relied on Author-ity (Torvik and Smalheiser 2021) and MapAffil datasets (Torvik 2021), which provide disambiguated author IDs and standardized affiliations for articles indexed in PubMed, the leading

²³ These measures are available in Protein Data Bank in Europe (PDBe). The PDB consists of a consortium of global partners, such as PDBe and RCSB PDB (the US branch). While each partner maintains its own website, they share the same underlying global data.

database of life science publications. As detailed in Appendix A.8, I used a combination of automated matching and extensive manual review to link PIs to PDB structures.

After obtaining the author IDs, I constructed three measures. First, the PI was classified as being from an elite institution in a given year if their affiliation appeared in the QS World University Rankings for Life Sciences and Medicine.²⁴ Second, as an alternative measure of pedigree, I tracked the cumulative number of citations the PI had accrued each year. Third, I assessed the PI’s prior experience in structural biology each year by calculating the cumulative share of structural biology–related keywords in their publication history.²⁵ This distinguishes PIs who are specialists in structural biology (those whose publications are primarily related to structural biology) versus PIs whose involvement in structural biology is only peripheral.

Table 6 reports the results. While there is no significant difference between bright and dark clusters in the number of structures solved by PIs from elite institutions, bright clusters saw a greater increase in structures contributed by non-elite PIs. Similar patterns hold when pedigree is measured using citations. Furthermore, bright clusters experienced a disproportionate increase in structures from PIs with less specialization in structural biology. These patterns are consistent with MR having a democratizing effect, lowering technical barriers that attracted entrants—PIs whose research agendas were not focused on structural biology and those with less elite background who may have faced higher costs due to limited resources that prevented them from entry.

Importantly, understanding who drove the increased activity in bright clusters helps clarify whether the increase in bright clusters after MR came at the expense of exploring dark clusters. Crowding out the exploration of dark clusters could be concerning since high-value proteins in dark clusters may become neglected, and the spillovers from solving the first structure in a dark cluster—which generates data that enables MR for other proteins in the cluster—would be forgone. The difference-in-differences approach, however, identifies only the *relative* increase in solved structures between bright and dark clusters, and this relative increase can represent either an overall expansion of structural biology research or a reallocation of efforts from dark to bright clusters. While I cannot separate this out perfectly, since the increase is disproportionately driven

²⁴ I applied the 2023 QS World University Rankings, as the ranking system was first introduced in 2004 and is not available for the earlier part of the sample period. One limitation of this measure is that it excludes elite non-university research institutions.

²⁵ PubMed’s curators annotate every publication with standardized keywords known as MeSH (Medical Subject Headings) terms. A MeSH term was classified as structural biology–related if it fell within the top 5% of MeSH terms that appear the most frequently in PDB publications. For each year in a PI’s publication history, I compiled all MeSH terms appearing across their publications and calculated the share corresponding to structural biology, providing a year-specific measure of structural biology specialization.

by non-elite PIs who had limited prior involvement in the field, this suggests that their post-MR activity reflects entry rather than reallocation. MR appears to have attracted new participants to structural biology, rather than diverting effort away from dark clusters.

7. Discussion

Using the setting of structural biology—which offers a unique window into the full knowledge landscape—I study the introduction of a data-extrapolation tool, MR, which facilitates structure determination by leveraging prior templates. MR reduced the cost of solving proteins in data-rich (bright) clusters, sparking an increase in the quantity of structures. While this did not translate into a corresponding increase in functional insights, the additional structures served as technical inputs to downstream research, were often well-executed, and attracted new entrants.

Broadly, these findings highlight that data-extrapolation technologies such as MR operate by lowering costs in nearby areas where data exists. While much of the prior literature on data-extrapolation tools focuses on the quality of the training data or the capability of the algorithms, this paper emphasizes that the very presence and absence of data dictates which opportunities are available for algorithmic tools at all. The knowledge landscape is often conceptualized as “rugged,” with peaks and valleys of opportunities (Kauffman 1993; Stuart and Podolny 1996; Levinthal 1997; Fleming and Sorenson 2004). The data used to navigate this terrain is unevenly generated, accumulating in areas that were historically important or tractable. Extrapolation technologies inherit this uneven coverage and may reinforce existing research trajectories, accelerating discovery where data is abundant, while offering limited traction in data-scarce regions.

One factor not examined in this paper that may further explain the impact of MR is the scope of its extrapolation. MR reduces the need for costly experimental phasing, but enables only limited extrapolation across closely related proteins, operating within narrowly defined technical neighborhoods. Data-extrapolation tools with sophisticated algorithmic capabilities that can span more disparate domains may yield larger cost reductions and spur higher-impact breakthroughs, from writing (Noy and Zhang 2023) to drug discovery (Lou and Wu 2021).

While MR operates within narrower technical bounds, the framework developed in this paper can help interpret the effects of more expansive data-extrapolation technologies. A timely example from structural biology is AlphaFold. In 2020, Google’s DeepMind team cracked a 50-year-old grand challenge in biology: to predict how a protein folds into its 3D structure from purely its sequence of amino acids. While MR helps with one part of experimental structure solving,

DeepMind’s AlphaFold algorithm bypasses the need for experiments at all. Celebrated as one of the most significant applications of AI, the DeepMind team was awarded the Nobel Prize in 2024.

Despite its breakthrough, however, AlphaFold also serves as a reminder of the limitations of data-extrapolation technologies. DeepMind’s claim that AlphaFold has produced enough structures to cover the “entire protein universe” (Walsh 2022) must be qualified with an important caveat: AlphaFold is limited by its training data, the PDB, and can only populate the protein structural space based on analogies to known PDB structures. In particular, protein structures often change in the presence of small-molecule drugs, yet the PDB contains limited data on drug-bound complexes, making it difficult for AlphaFold to model such interaction (Callaway 2022).²⁶

Facing these data challenges, a consortium of pharmaceutical firms announced in 2025 that they plan to build their own version of AlphaFold, trained on their internal database of proprietary structures (Callaway 2025). The PDB, built over decades by academic scientists, made the data freely available to generate spillovers in exchange for priority. This enabled the remarkable development of AlphaFold, whose own source code has also been made public. In contrast, the pharmaceutical consortium reveals an alternative approach to public data infrastructure: keeping data private altogether. As a result, one of the most critical applications of AlphaFold—drug discovery—will likely occur solely within private firms.

AlphaFold therefore highlights that when assessing the impact of data-extrapolation tools, it is essential to consider not only where data is generated, but also who owns it. This concern is particularly salient as the frontier of AI increasingly resides in industry, where firms can leverage proprietary datasets for strategic advantage (Ahmed, Wahed, and Thompson 2023). For example, for firms that produce products and services based on data, early entrants may use their control over data to outcompete rivals (Bessen et al. 2022).

Finally, AlphaFold underscores the importance of comprehensive, exploratory data generation—“mapping for its own sake” (Nagaraj and Stern 2020). MR enabled structures that may have been initially dismissed as incremental, and initiatives whose sole goals were to solve as many structures as possible were often criticized for lacking a hypothesis-driven agenda (Petsko 2007; Zhuo 2023). Yet these initiatives illustrate how data generated without apparent scientific value may later prove pivotal for extrapolation. MR’s own role in increasing the number of solved structures helped lay the groundwork for AlphaFold by expanding the pool of training data. As

²⁶ Lou and Wu (2021) have found that AI is less useful for developing drugs that are radically novel and have no known mechanisms of action. In addition, a recent paper investigates how AlphaFold changed the organizational structure of academic labs in computational biology (Cavalli 2022).

data-extrapolation tools become increasingly central to innovation, navigating the rugged data landscape requires understanding how prior data generation efforts enable or constrain future discoveries.

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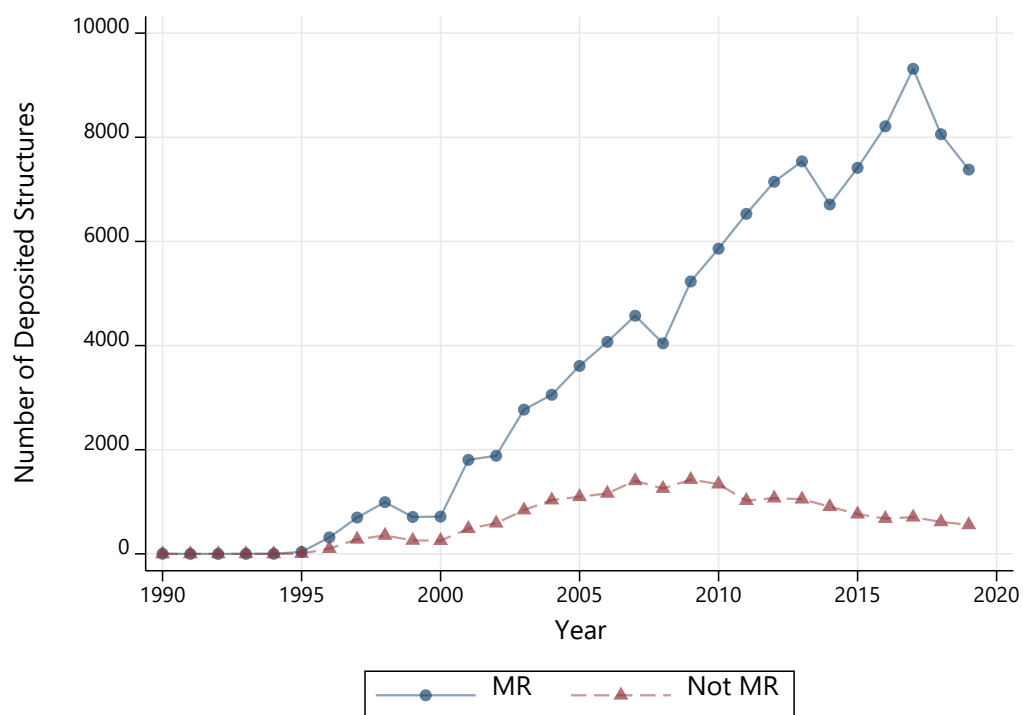
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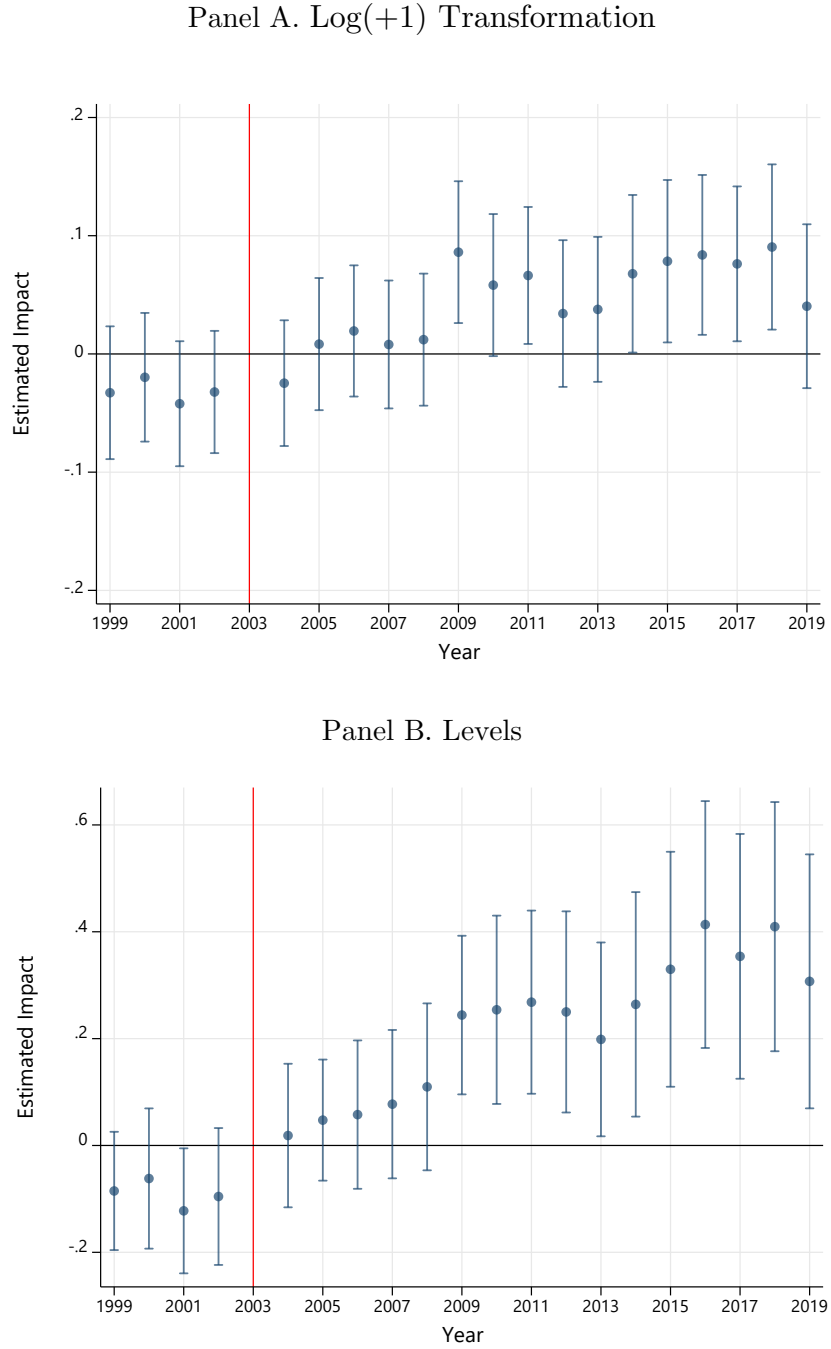
Figures & Tables

FIGURE 1. NUMBER OF STRUCTURES SOLVED BY MOLECULAR REPLACEMENT



NOTES: This figure plots the number of X-ray crystallography structures in the Protein Data Bank that were solved by molecular replacement (MR) vs. non-MR methods.

FIGURE 2. EVENT STUDY: IMPACT OF MR ON NUMBER OF SOLVED STRUCTURES



NOTES: This figure shows the impact of MR on the number of solved structures. The figure plots the coefficients and 95% confidence intervals from estimating an event studies version of Equation 1 that replaces the pooled $PostMR_t$ indicator with separate indicators for every year before and after the arrival of MR. The outcome is the total annual number of solved structures in a cluster; Panel A reports the outcome after Log(+1) transformation, while Panel B reports the outcome in levels. The unit of analysis is a cluster $\times$ year, and the sample consists of 6,942 clusters, which translates to 145,782 cluster-years.

TABLE 1. SUMMARY STATISTICS

	Bright Clusters		Dark Clusters	
	Mean	SD	Mean	SD
Cluster Discovery Year	1983.5	7.7	1993.1	5.1
Cluster Size	39.1	80.5	7.6	14.7
Features Related to Structure Solving Difficulty				
Share of Disordered Proteins	20%	36%	38%	46%
Share of Membrane Proteins	2%	14%	7%	25%
Share of Proteins with Compositional Bias	1%	8%	4%	18%
Sequence Length	512.0	1,434.9	634.8	676.0
Features Related to Biological Importance				
Share of Human Proteins	18%	15%	36%	24%
Share of Proteins with Known Function	67%	32%	42%	40%
Share of Proteins with Disease Relevance	5%	12%	4%	13%
N of Publications per Protein	6.9	10.2	4.2	8.6
N of Approved Drugs per Protein	0.2	0.7	0.1	0.5
N of Structures Solved per Year	1.3	3.7	0.1	0.5
N of Clusters	653		6,289	

NOTES: This table provides the summary characteristics of clusters when MR was introduced in 2003. The sample consists of 6,942 clusters, of which 653 are classified as “bright” (i.e., had at least one structure by 1998) and 6,289 are classified as “dark.” Cluster discovery year is the earliest recorded existence of a protein in the cluster, based on either its initial entry in major sequence databases (such as UniProt, the European Molecular Biology Laboratory database, and the DNA Data Bank of Japan) or its earliest mention in a publication, as documented by UniProt. Proteins are categorized as disordered, membrane, or compositionally biased when at least a quarter of their amino acid sequence is associated with these properties. Information on publications, functional annotation, and disease relevance as of 2003 was parsed from UniProt, while drug-related information was obtained from DrugBank.

TABLE 2. IMPACT OF MR ON NUMBER OF SOLVED STRUCTURES

VARIABLES	(1) Log(+1) N of Structures	(2) Log(+1) N of Structures	(3) Levels N of Structures	(4) Levels N of Structures
Post-MR $\times$ Bright	0.072*** (0.018)	0.042** (0.018)	0.298*** (0.051)	0.223*** (0.042)
R-squared	0.471	0.474	0.390	0.394
Calendar year FE	YES	YES	YES	YES
Cluster FE	YES	YES	YES	YES
Cluster size FE	NO	YES	NO	YES
Cluster age FE	NO	YES	NO	YES
N of structures (mean)	0.400	0.400	0.400	0.400
N of structures (SD)	2.450	2.450	2.450	2.450
N of clusters	6,942	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782	145,782

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The outcome variable is the total annual number of solved structures in a cluster, reported after Log(+1) transformation (Columns 1-2) or in levels scaled by the standard deviation (Columns 3-4). Means and standard deviations of the outcomes are reported in levels. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar year and cluster fixed effects; Columns 2 and 4 additionally control for time-varying cluster size and cluster age. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 3. IMPACT OF MR ON FUNCTIONAL INSIGHTS

VARIABLES	(1) Unpublished Structures	(2) Published Structures	(3) Published Structures <i>Not Cited by Patent</i>	(4) Published Structures <i>Cited by Patent</i>	(5) Published Structures <i>Not Fxn Annotated</i>	(6) Published Structures <i>Fxn Annotated</i>
Post-MR $\times$ Bright	0.082*** (0.008)	0.041** (0.017)	0.069*** (0.017)	-0.018* (0.009)	0.041** (0.017)	0.005 (0.004)
R-squared	0.213	0.465	0.416	0.318	0.473	0.123
N of structures (mean)	0.0500	0.340	0.260	0.080	0.310	0.030
N of structures (SD)	1.010	2.090	1.760	0.810	2.040	0.340
N of clusters	6,942	6,942	6,942	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782	145,782	145,782	145,782

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures by whether the structure elucidated functional insights. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The outcomes of all columns are the annual number of solved structures in a cluster, with Log(+1) transformation. Means and standard deviations of the outcomes are reported in levels. Columns 1 and 2 parallel the outcome in Column 1 of Table 2 but decompose the number of solved structures by whether they were cited in a scientific article. Columns 3 and 4 decompose the number of solved structures by whether they were cited by a patent within 5 years of publication. Columns 5 and 6 decompose the number of solved structures by whether they were cited by the functional summary section of Swiss-Prot as of January 2024. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 4. IMPACT OF MR ON EXECUTION

VARIABLES	(1) <i>Resolution</i> Bottom Tercile	(2) <i>Resolution</i> Middle Tercile	(3) <i>Resolution</i> Top Tercile	(4) <i>R-Free</i> Bottom Tercile	(5) <i>R-Free</i> Middle Tercile	(6) <i>R-Free</i> Top Tercile
Post-MR $\times$ Bright	0.009 (0.011)	0.072*** (0.012)	0.092*** (0.012)	0.043*** (0.011)	0.067*** (0.012)	0.089*** (0.012)
R-squared	0.366	0.404	0.423	0.376	0.394	0.420
N of structures (mean)	0.130	0.130	0.110	0.120	0.120	0.110
N of structures (SD)	0.700	0.940	1.460	0.800	1.060	1.060
N of clusters	6,942	6,942	6,942	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782	145,782	145,782	145,782

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures at different terciles of execution level (a structure's level of execution can be defined in terms of resolution and R-free values). The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The outcomes of all columns are the annual number of solved structures in a cluster, with Log(+1) transformation. Means and standard deviations of the outcomes are reported in levels. Columns 1-3 parallel Column 1 in Table 2 but decompose the outcome into the number of solved structures by terciles based on their resolution. Columns 4-6 similarly decompose the number of solved structures by terciles based on their R-free values. For both resolution and R-free, lower values indicate higher quality; I construct terciles such that higher terciles correspond to higher quality. The treatment variable "Bright" is defined as clusters that had at least one structure by 1998, while "Post-MR" includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 5. IMPACT OF MR ON CITATIONS AND MENTIONS

	(1)	(2)	(3)	(4)	(5)
	Published Structures <i>Citations</i>			Published Structures <i>Mentions</i>	
VARIABLES	Bottom Tercile	Middle Tercile	Top Tercile	Below Median	Above Median
Post-MR $\times$ Bright	0.046*** (0.012)	0.033*** (0.012)	0.016 (0.010)	0.025* (0.015)	0.059*** (0.012)
R-squared	0.363	0.309	0.332	0.427	0.385
N of structures (mean)	0.110	0.110	0.120	0.190	0.150
N of structures (SD)	0.900	0.870	1.190	1.640	0.800
N of clusters	6,942	6,942	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782	145,782	145,782

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures at different levels of citations and mentions. A structure’s citation impact is measured as the mean annual number of citations within 5 years of publication. In addition to being formally cited, a structure can be “mentioned” in the text of downstream articles; a structure’s mention impact is measured as the mean annual number of mentions within 5 years of publication. The unit of analysis is a cluster $\times$ year. The outcomes of all columns are the annual number of solved structures in a cluster, with Log(+1) transformation. Means and standard deviations of the outcomes are reported in levels. Columns 1-3 parallel Column 1 in Table 2 but decompose the outcome into the number of solved structures by terciles of citation impact. Columns 4 and 5 decompose the number of solved structures by whether they fall below or above the median level of mentions. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 6. IMPACT OF MR BY THE CHARACTERISTICS OF THE PRINCIPAL INVESTIGATOR

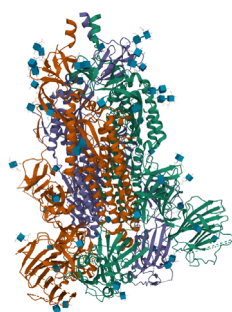
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pedigree					Prior Structural Biology Experience		
	Affiliation		Prior N of Citations			Share of Structural Biology Keywords in Prior Papers		
VARIABLES	Not Top 20	Top 20	Bottom Tercile	Middle Tercile	Top Tercile	Bottom Tercile	Middle Tercile	Top Tercile
Post-MR $\times$ Bright	0.073*** (0.016)	0.006 (0.006)	0.050*** (0.013)	0.053*** (0.012)	0.026** (0.012)	0.058*** (0.015)	0.047*** (0.012)	0.018* (0.010)
R-squared	0.448	0.226	0.392	0.322	0.317	0.445	0.313	0.267
N of structures (mean)	0.250	0.050	0.120	0.120	0.110	0.140	0.120	0.090
N of structures (SD)	1.740	1.000	0.970	1.500	0.840	1.140	1.240	1.060
N of clusters	6,942	6,942	6,942	6,942	6,942	6,942	6,942	6,942
N of cluster-years	124,956	124,956	124,956	124,956	124,956	124,956	124,956	124,956

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures, decomposed by the characteristics of the Principal Investigator (PI). The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2016, ending in 2016 due to data availability. The outcomes of all columns are the annual number of solved structures in a cluster, with Log(+1) transformation. Means and standard deviations of the outcomes are reported in levels. Columns 1–2 parallel Column 1 of Table 2 but distinguish between number of structures solved by PIs affiliated with institutions ranked in the top 20 of the QS World University Rankings and those at institutions outside the top 20. Columns 3–5 decompose the number of solved structures by terciles of the PI’s prior total citation count. Columns 6–8 decompose the number of solved structures by the PI’s prior structural biology experience, using terciles based on the share of structural biology-related MeSH keywords in the PI’s publication history in a given year. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

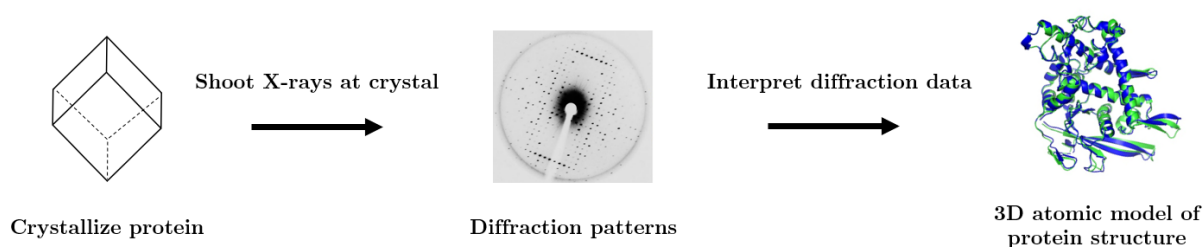
Appendix Figures & Tables

APPENDIX FIGURE 1. STRUCTURAL BIOLOGY

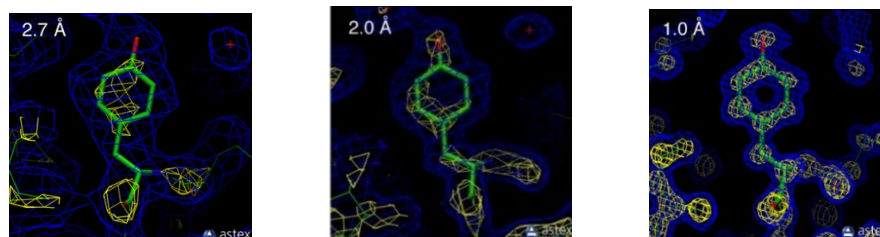
A. Structure of the SARS-CoV-2 Spike Glycoprotein



B. Steps of Crystallography



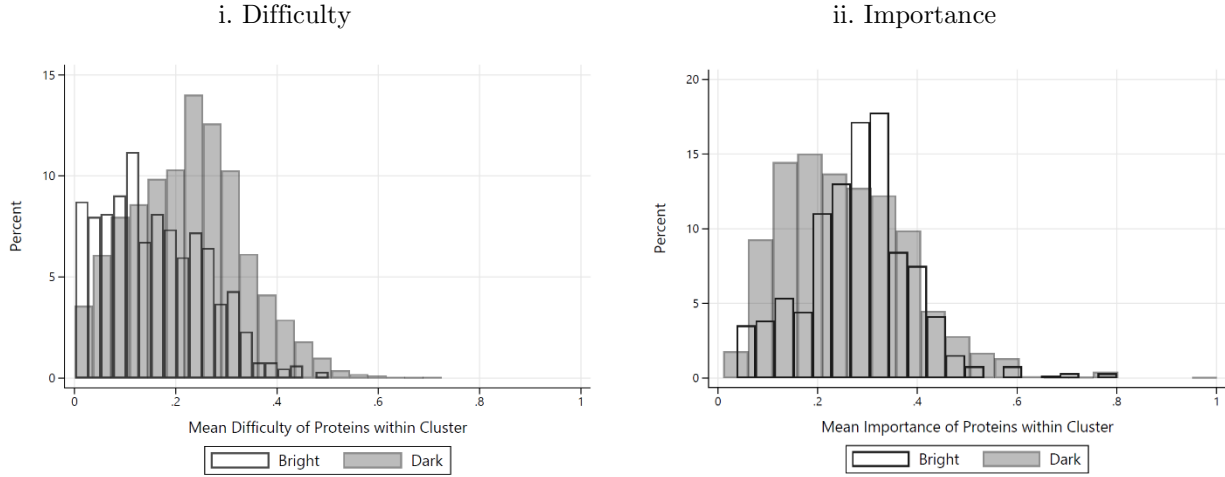
C. Resolution



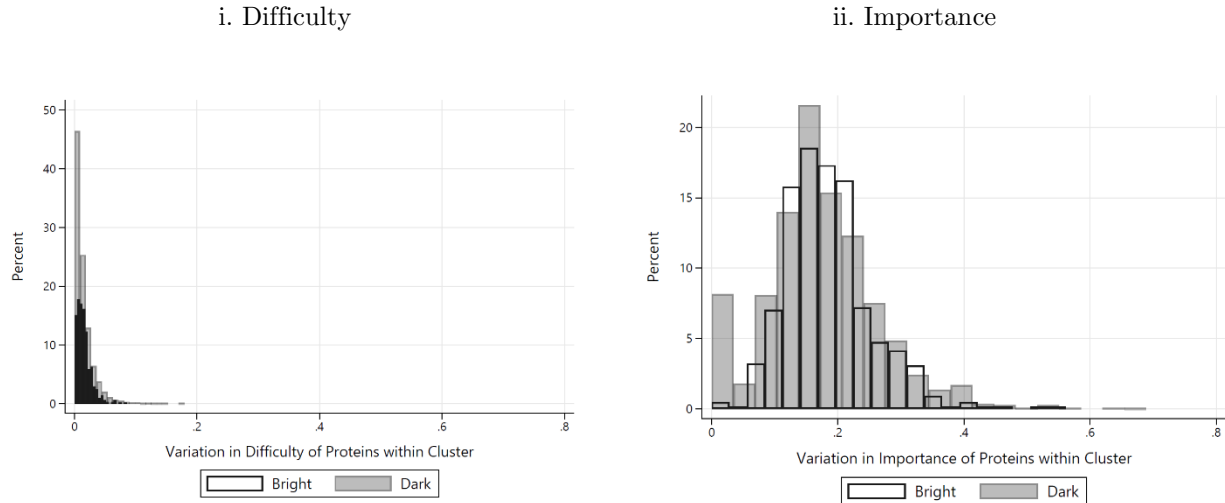
NOTES: Panel A shows the structure of a spike protein on the surface of the coronavirus (PDB entry 6VYB; source: <https://www.rcsb.org/structure/6VYB>). Panel B shows the three main steps of crystallography; this paper focuses on the automation of solving the “phase problem” that occurs during the interpretation of the diffraction data. Panel C shows an example of the electron density map behind the structure of tyrosine 103 from myoglobin, at three different resolutions (source: <https://pdb101.rcsb.org/learn/guide-to-understanding-pdb-data/resolution>).

APPENDIX FIGURE 2. CLUSTER-LEVEL DISTRIBUTIONS OF DIFFICULTY AND IMPORTANCE

Panel A. Distribution of *Mean* Difficulty and Importance of Proteins within Cluster



Panel B. Distribution of *Variation* in Difficulty and Importance of Proteins within Cluster



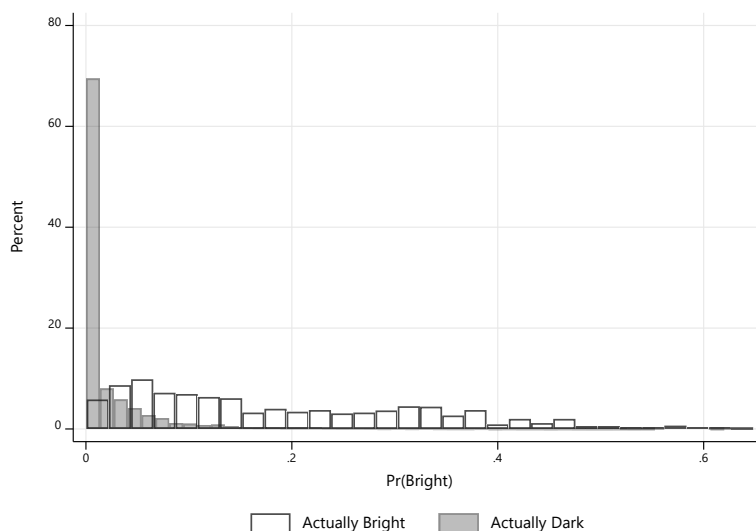
NOTES: These figures plot the distribution of cluster-level difficulty and importance scores among 6,942 clusters as of 2003 when MR arrived. Panel A plots the mean scores of proteins within each cluster, while Panel B plots the variation in scores (measured by the standard deviation) of proteins within each cluster. The difficulty score of each protein is calculated as the average of four features: (i) the share of amino acids classified as membrane-associated, (ii) the share of amino acids classified as intrinsically disordered, (iii) the share of amino acids exhibiting compositional bias, and (iv) the percentile (scaled 0–1) of sequence length. The importance score of each protein is calculated as the average of five features: (i) whether the protein is associated with a disease, (ii) whether the protein has a known function, (iii) whether the protein is targeted by an approved drug, (iv) whether the protein is from human, and (v) the percentile (scaled 0–1) of number of publications written about the protein. Protein-level scores are then aggregated to the cluster level by taking the mean and standard deviation across proteins in each cluster.

APPENDIX FIGURE 3. PERFORMANCE AND DISTRIBUTION OF PREDICTED BRIGHTNESS

A. Performance of Lasso Logit

Performance Metric	Out-of-Sample (Cross-Validated)	In-Sample (Fitted Model)
ROC AUC	0.91	0.93
PR AUC	0.24	0.33
Log Loss	0.09	0.08
N of Features Entered	143	
N of Features Selected	28	

B. Distribution of Predicted Brightness



NOTES: This figure provides details on the performance and distribution of the predicted brightness measure. The predicted brightness measure was estimated using a Lasso Logit model, which predicts whether a protein was structurally characterized by 1998 based on pre-period biological and technical feasibility features (see Appendix A.6 for details). Panel A reports model performance based on both out-of-sample (cross-validated) and in-sample (fitted) predictions, including Receiver Operating Characteristic Area Under Curve (ROC AUC), Precision-Recall Area Under Curve (PR AUC), and log loss. The Lasso Logit model selected 28 features out of 143 candidates. Panel B plots the distribution of the resulting predicted brightness by whether the protein was actually bright (i.e., had a structure by 1998) or dark (i.e., did not have a structure by 1998). The sample consists of 42,547 proteins that were discovered as of 1998.

APPENDIX TABLE 1. IMPACT OF MR ON NUMBER OF SOLVED STRUCTURES, INVERSE
HYPERBOLIC SINE TRANSFORMATION

VARIABLES	(1)	(2)
	IHS N of Structures	IHS N of Structures
Post-MR $\times$ Bright	0.084*** (0.022)	0.047** (0.022)
R-squared	0.464	0.468
Calendar year FE	YES	YES
Cluster FE	YES	YES
Cluster size FE	NO	YES
Cluster age FE	NO	YES
N of structures (mean)	0.400	0.400
N of structures (SD)	2.450	2.450
N of clusters	6,942	6,942
N of cluster-years	145,782	145,782

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The outcome variable is the total annual number of solved structures in a cluster, reported after inverse hyperbolic sine transformation. Means and standard deviations of the outcomes are reported in levels. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar year and cluster fixed effects; Columns 2 and 4 additionally control for time-varying cluster size and cluster age. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

APPENDIX TABLE 2. IMPACT OF MR ON MR VS. NON-MR STRUCTURES

VARIABLES	(1) All Structures	(2) MR Structures	(3) Non-MR Structures
Post-MR $\times$ Bright	0.072*** (0.018)	0.143*** (0.017)	-0.007* (0.004)
R-squared	0.471	0.475	0.102
N of structures (mean)	0.400	0.290	0.0200
N of structures (SD)	2.450	2.170	0.190
N of clusters	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782

NOTES: This table parallels Column 1 from Table 2. The table reports results from estimating Equation 1 and shows the impact of MR on the total number of solved structures (Column 1) and decomposes this into number of MR structures (Column 2) and non-MR structures (Column 3). All of the outcomes are Log(+1) transformed. Means and standard deviations of the outcomes are reported in levels. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

APPENDIX TABLE 3. IMPACT OF MR, SPLITTING BRIGHT CLUSTERS

	(1)	(2)	(3)
	Dark vs. All Bright Clusters	Dark vs. Bright Clusters with 1 Structure	Bright Clusters with 1 Structure vs. Bright Clusters with >1 Structures
Post-MR $\times$ Bright (1 or more structure)	0.072*** (0.018)	0.068*** (0.022)	
Post-MR $\times$ Bright (more than 1 structure)			0.006 (0.034)
R-squared	0.471	0.324	0.592
N of structures (mean)	0.400	0.250	1.980
N of structures (SD)	2.450	1.830	5.540
N of clusters	6,942	6,549	653
N of cluster-years	145,782	137,529	13,713

NOTES: Column 1 of this table parallels Column 1 of Table 2 and shows the impact of MR on the total number of solved structures in the full sample. Column 2 investigates the impact of MR on the sample of dark clusters and bright clusters with just 1 structure solved by 1998; the treatment variable “Bright” is defined as clusters that had just one structure by 1998. Column 3 investigates the impact of MR on the sample of bright clusters with 1 or more structures solved by 1998; the treatment variable “Bright” is defined as clusters that had more than 1 structure by 1998. All of the outcomes are Log(+1) transformed. Means and standard deviations of the outcomes are reported in levels. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** p<0.01, ** p<0.05, * p<0.1.

APPENDIX TABLE 4. IMPACT OF MR ON FUNCTIONAL INSIGHTS: HARD VS. EASY CLUSTERS

VARIABLES	Hard Clusters						Easy Clusters					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Unpublished Structures	Published Structures	Published Structures <i>Not Cited by Patent</i>	Published Structures <i>Cited by Patent</i>	Published Structures <i>Not Fxn Annotated</i>	Published Structures <i>Fxn Annotated</i>	Unpublished Structures	Published Structures	Published Structures <i>Not Cited by Patent</i>	Published Structures <i>Cited by Patent</i>	Published Structures <i>Not Fxn Annotated</i>	Published Structures <i>Fxn Annotated</i>
Post-MR $\times$ Bright	0.057*** (0.014)	0.093*** (0.032)	0.117*** (0.030)	0.005 (0.019)	0.090*** (0.032)	0.006 (0.006)	0.092*** (0.010)	0.026 (0.02)	0.053*** (0.020)	-0.024** (0.011)	0.027 (0.020)	0.004 (0.006)
R-squared	0.158	0.401	0.328	0.318	0.406	0.112	0.246	0.504	0.464	0.318	0.513	0.130
N of structures (mean)	0.04	0.260	0.190	0.0700	0.240	0.0200	0.0700	0.42	0.330	0.0900	0.390	0.0400
N of structures (SD)	1.24	1.960	1.630	0.800	1.920	0.290	0.700	2.21	1.870	0.810	2.140	0.370
N of clusters	3,471	3,471	3,471	3,471	3,471	3,471	3,471	3,471	3,471	3,471	3,471	3,471
N of bright clusters	185	185	185	185	185	185	468	468	468	468	468	468
N of dark clusters	3,286	3,286	3,286	3,286	3,286	3,286	3,003	3,003	3,003	3,003	3,003	3,003
N of cluster-years	72,891	72,891	72,891	72,891	72,891	72,891	72,891	72,891	72,891	72,891	72,891	72,891

NOTES: This table parallels Table 3 but splits the sample by cluster-level difficulty. The difficulty score of each protein is calculated as the average of four features: (i) the share of amino acids classified as membrane-associated, (ii) the share of amino acids classified as intrinsically disordered, (iii) the share of amino acids exhibiting compositional bias, and (iv) the percentile (scaled 0–1) of sequence length. Protein-level scores are then averaged to the cluster level. Clusters with above-median scores are classified as “hard,” while those below the median are classified as “easy.” See Appendix A.5 for more details.

APPENDIX TABLE 5. IMPACT OF MR ON JOURNAL IMPACT AND SPECIALIZATION

	(1)	(2)	(3)	(4)	(5)	(6)
	Journal Impact Factor			Journal Specialization in Structural Biology		
VARIABLES	Bottom Tercile	Middle Tercile	Top Tercile	Low Specialization Tercile	Moderate Specialization Tercile	High Specialization Tercile
Post-MR $\times$ Bright	0.056*** (0.012)	0.021* (0.011)	0.014 (0.010)	0.016 (0.011)	0.004 (0.012)	0.058*** (0.010)
R-squared	0.406	0.302	0.287	0.345	0.326	0.269
N of structures (mean)	0.130	0.110	0.110	0.130	0.110	0.0900
N of structures (SD)	1.070	0.890	1.080	1.230	0.930	0.720
N of clusters	6,942	6,942	6,942	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782	145,782	145,782	145,782

NOTES: This table reports results from estimating Equation 1 and shows the impact of MR on the number of solved structures at different terciles of journal impact and specialization. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The outcomes of all columns are the annual number of solved structures in a cluster, with Log(+1) transformation. Means and standard deviations of the outcomes are reported in levels. Columns 1-3 decompose the number of solved structures by terciles of journal impact factor. Columns 4-6 decompose the number of solved structures in journals that are high or low in structural biology specialization. Journal specialization is measured as the average share of structural biology-related keywords per article, calculated at the journal level over the sample period; journals are then classified into terciles based on this measure. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

APPENDIX TABLE 6. IMPACT OF MR ON NUMBER OF SOLVED STRUCTURES WITH PREDICTED BRIGHTNESS

A. Inclusion of Post-MR $\times$ Predicted Bright

VARIABLES	(1)	(2)	(3)	(4)
	Log(+1) N of Structures	Log(+1) N of Structures	Levels N of Structures	Levels N of Structures
Post-MR $\times$ Bright	0.065*** (0.019)	0.046*** (0.018)	0.276*** (0.052)	0.224*** (0.043)
Post-MR $\times$ Predicted Bright	0.168 (0.127)	-0.173 (0.132)	0.542** (0.251)	-0.040 (0.271)
R-squared	0.471	0.474	0.390	0.394
Calendar year FE	YES	YES	YES	YES
Cluster FE	YES	YES	YES	YES
Cluster size FE	NO	YES	NO	YES
Cluster age FE	NO	YES	NO	YES
N of structures (mean)	0.400	0.400	0.400	0.400
N of structures (SD)	2.450	2.450	2.450	2.450
N of clusters	6,942	6,942	6,942	6,942
N of cluster-years	145,782	145,782	145,782	145,782

B. Use of Propensity Score Matching

VARIABLES	(1)	(2)	(3)	(4)
	Log(+1) N of Structures	Log(+1) N of Structures	Levels N of Structures	Levels N of Structures
Post-MR X Bright	0.055*** (0.021)	0.047** (0.021)	0.262*** (0.053)	0.199*** (0.042)
R-squared	0.568	0.574	0.536	0.548
Calendar year FE	YES	YES	YES	YES
Cluster FE	YES	YES	YES	YES
Cluster size FE	NO	YES	NO	YES
Cluster age FE	NO	YES	NO	YES
N of structures (mean)	1.190	1.190	1.190	1.190
N of structures (SD)	4.220	4.220	4.220	4.220
N of clusters	1,306	1,306	1,306	1,306
N of cluster-years	27,426	27,426	27,426	27,426

NOTES: These tables parallel Table 2 and show the impact of MR on the number of solved structures, controlling for predicted brightness. The predicted brightness measure was estimated using a Lasso Logit model, predicting whether a protein was structurally characterized by 1998 based on pre-period features on biological importance and technical feasibility. These scores were then averaged to the cluster level (see Appendix A.6 for details). In Panel A, the “Post-MR $\times$ Predicted Bright” term was added to Equation 1. In Panel B, one-to-one nearest-neighbor propensity score matching was implemented on the predicted brightness measure; unmatched dark clusters are dropped from the sample. The unit of analysis is a cluster $\times$ year, and the panel spans from 1999-2019. The outcome variable is the total annual number of solved structures in a cluster, reported after Log(+1) transformation (Columns 1-2) or in levels scaled by the standard deviation (Columns 3-4). Means and standard deviations of the outcomes are reported in levels. The treatment variable “Bright” is defined as clusters that had at least one structure by 1998, while “Post-MR” includes years 2004 and onwards. All columns include calendar-year and cluster fixed effects. Columns 2 and 4 additionally control for time-varying cluster size and cluster age. Standard errors are clustered at the cluster level. Statistical significance is indicated as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

APPENDIX TABLE 7. IMPACT OF MR ON NUMBER OF SOLVED STRUCTURES: ALTERNATIVE SAMPLE CONSTRUCTION

A. Unbalanced Panel

	(1)	(2)	(3)	(4)
	Log(+1)	Log(+1)	Levels	Levels
VARIABLES	N of Structures	N of Structures	N of Structures	N of Structures
Post-MR $\times$ Bright	0.088*** (0.018)	0.046*** (0.017)	0.396*** (0.063)	0.279*** (0.051)
R-squared	0.464	0.468	0.379	0.383
Calendar year FE	YES	YES	YES	YES
Cluster FE	YES	YES	YES	YES
Cluster size FE	NO	YES	NO	YES
Cluster age FE	NO	YES	NO	YES
N of structures (mean)	0.250	0.250	0.250	0.250
N of structures (SD)	1.970	1.970	1.970	1.970
N of clusters	12,292	12,292	12,292	12,292
N of cluster-years	248,677	248,677	248,677	248,677

B. UniProt Similarity Families

	(1)	(2)	(3)	(4)
	Log(+1)	Log(+1)	Levels	Levels
VARIABLES	N of Structures	N of Structures	N of Structures	N of Structures
Post-MR $\times$ Bright	0.131*** (0.020)	0.069*** (0.020)	0.392*** (0.055)	0.282*** (0.046)
R-squared	0.542	0.545	0.453	0.457
Calendar year FE	YES	YES	YES	YES
Family FE	YES	YES	YES	YES
Family size FE	NO	YES	NO	YES
Family age FE	NO	YES	NO	YES
N of structures (mean)	0.550	0.550	0.550	0.550
N of structures (SD)	3.360	3.360	3.360	3.360
N of families	6,066	6,066	6,066	6,066
N of family-years	127,386	127,386	127,386	127,386

NOTES: These tables parallel Table 2 and show the impact of MR on the number of solved structures under alternative sample constructions. Panel A presents results using an unbalanced panel that includes clusters discovered by 2003. Panel B reports results based on an alternative cluster definition using UniProt similarity families. See Appendix A.7 for more details.

Appendix A. Data and Additional Analyses

A.1 UniProt/Swiss-Prot

The Universal Protein Resource Knowledgebase (UniProt) is a comprehensive database of known proteins. A protein is composed of a sequence of organic compounds called amino acids. Information for making a protein is stored in a gene’s DNA, and by translating the DNA sequence of a gene, scientists can determine the protein’s existence and the sequence of amino acids that appear in the protein. Protein sequences in UniProt are thus sourced by translating genes from major genome sequence databases.

UniProt is divided into two parts: Swiss-Prot (manually reviewed) and TrEMBL (computationally reviewed). Created in 1986, the Swiss-Prot database is extensively reviewed, maintained, and annotated by experts based on experimental results and literature review. As of October 2020, Swiss-Prot contains 563,552 protein entries. In contrast, TrEMBL was created in 1996 and houses computationally annotated protein entries. Once a protein from TrEMBL becomes manually reviewed, it is removed from TrEMBL and enters Swiss-Prot. TrEMBL was established in recognition that manual curation efforts cannot keep pace with the increased number of protein sequences resulting from genome sequence projects and contains nearly two hundred million entries.

To define the complete set of proteins at risk of being structurally characterized, I follow (Perdigão et al. 2015)—a bioinformatics paper that descriptively mapped which proteins’ structures have been determined—and focus on the proteins in the Swiss-Prot database. While smaller than TrEMBL, using the Swiss-Prot database has several advantages. First, Swiss-Prot is one of the best datasets of proteins whose existence is experimentally proven (Perdigão et al. 2015); TrEMBL primarily contains proteins whose existence is only predicted. Second, since Swiss-Prot primarily includes well-characterized proteins, focusing on Swiss-Prot ensures that I examine proteins with a comparable baseline level of annotation and visibility to structural biologists, rather than unreviewed entries that may not correspond to real proteins. Third, Swiss-Prot’s expertly curated annotation provides rich descriptions of each protein, including its function, clinical impact, and sequence features, which allows me to develop difficulty and importance scores for each protein, as well as a “predicted brightness” measure, as described in Appendix A.5 and A.6.

A.2 Linking Swiss-Prot to the Protein Data Bank

The PDB provides crosswalks to Swiss-Prot, which can be used to observe which proteins in Swiss-Prot have had their structures characterized in the PDB. However, the level of the crosswalk between an entry in the PDB and an entry in Swiss-Prot is not a many-to-one crosswalk as one might expect (a many-to-one, since a protein in Swiss-Prot can have its structure solved multiple times), but rather a many-to-many crosswalk (i.e., a single protein structure in the PDB can also be linked to multiple Swiss-Prot entries). This is because in the PDB, large protein structures can be composed of discrete regions called “entities”; the crosswalk between the PDB and Swiss-Prot is at this entity level. Approximately 80% of the structures in the PDB are composed of a single entity, while the remaining 20% have multiple entities and therefore linked to multiple Swiss-Prot entries. If a single protein structure from the PDB links to multiple Swiss-Prot entries, I split the protein structure into fractions based on the percentage of amino acids each Swiss-Prot entry contributes to the protein structure to avoid double counting a structure.

A.3 MMseqs2

MMseqs2 is a software package that clusters databases of proteins and can be downloaded at <https://github.com/soedinglab/MMseqs2> (Steinegger and Söding 2018; Hauser, Steinegger, and Söding 2016). MMseqs2 uses a greedy set cover algorithm and aims to create the fewest number of mutually exclusive clusters, given a set of proteins at a user-specified sequence similarity. In this paper, I chose the threshold of 30% sequence identity, given that MR will likely be successful if the template and the target proteins share at least 30% sequence identity.²⁷ If the identity falls below 30%, MR will be usually challenging, if at all possible, to implement (Schmidberger et al. 2010; Phenix 2022). The algorithm takes the following steps:

1. MMseqs2 first computes all pairwise sequence identities between proteins in Swiss-Prot
2. MMseqs2 chooses a “representative” sequence, which is the protein with the highest number of neighbors that share at least 30% sequence identity
3. MMseqs2 forms the first cluster with this representative sequence and all of its neighbors
4. MMseqs2 then looks at the remaining sequences and chooses the next representative sequence with the highest number of neighbors

²⁷ I also restricted the search such that the focal protein and the candidate protein shares at 80% coverage in their alignment in terms of sequence length.

5. MMseqs2 iterates through Steps 2-4 until all sequences belong in a cluster

This ensures that each member of a cluster shares at least 30% sequence identity with the representative sequence of the cluster.²⁸ MMseqs2 is used by both Swiss-Prot and the PDB to cluster similar proteins.

A.4 Sample Construction

Using the MMseqs2 algorithm, I grouped all 563,552 proteins in Swiss-Prot into 73,956 mutually exclusive clusters, using 30% sequence identity threshold.

Restricting to clusters with at least one human protein: I restricted the sample to clusters with at least one human protein ($n = 13,145$ clusters, which is equivalent to 158,623 proteins). There are two reasons for this restriction. First, restricting to clusters with at least one human protein ensures that all of the clusters in the final sample have a minimum baseline level of biological importance; one of the main goals of structural biology is to understand human biological processes and thus structural biologists are especially interested in proteins from humans (and their similarity neighbors). Second, focusing on human proteins mitigates the concern of growing cluster size. The total number of possible proteins in the universe is essentially infinite,²⁹ and new protein sequences are continuously discovered. However, all human proteins have been discovered by the early 2000s when the human genome project was completed; since one gene encodes one protein, and humans have approximately 20,000 genes, they also have 20,000 proteins.³⁰ Since the number of newly discovered human proteins have plateaued since the early 2000s when MR arrived, this alleviates the concern of whether clusters with human proteins are

²⁸ A caveat is that while it is likely that all possible pairs of sequences within the cluster also share at least 30% sequence similarity with each other (since they are all similar to the representative sequence), this is not guaranteed. Mirdita et al. (2017) performed a cluster quality check that mitigates this concern; the authors computed the mean sequence identity among all possible pairs of sequences in a cluster and found that MMseqs2 indeed yielded clusters where all possible pairs of sequences shared on average >30% sequence similarity.

²⁹ Given that there are 20 different amino acids and an average protein has a sequence length of 200 amino acids, this amounts to 20^{200} possible proteins, which is larger than the number of electrons in the universe (Koonin, Wolf, and Karev 2002).

³⁰ This is called the “one gene, one protein” rule, which contributed to the 1941 Nobel Prize in Medicine. As explained in Section 4.1, by translating the DNA sequence of a gene, scientists can determine the protein’s existence, and the sequence of amino acids that will appear in the final protein. Recently, the “one gene, one protein” rule has been challenged, as one gene may produce multiple proteins through, for instance, alternative splicing. Nonetheless, this paper follows the “one gene, one protein” rule since Swiss-Prot provides a non-redundant set of proteins, in that all proteins that are encoded by one gene in a species is folded into a single entry (including alternative splicing isoforms).

getting more structures due to MR or because there are simply more human proteins being discovered. To additionally address the concern of changing cluster size, I have specifications that control for time-varying cluster size.

Restricting to clusters born on or before 1998: For each cluster, I compute its discovery year by taking the earliest discovery year among the proteins in the cluster.³¹ Since my panel starts in 1999, I kept clusters that were born on or before 1998. This led to my final sample of 6,942 clusters (which is equivalent to 136,965 proteins).

A.5 Difficulty and Importance of Clusters

This section describes how cluster-level difficulty and importance scores were constructed. Difficulty features are time-invariant, as they reflect the physicochemical properties of the protein. Importance features are measured as of 2003, the year MR was introduced, and the analysis focuses on proteins that existed by that time. I first compute composite scores at the protein level and then aggregate them to the cluster level.

Below are the features related to technical difficulty of solving a protein:

- Membrane: what % of amino acids of the protein are in membrane regions?
- Disorder: what % of amino acids of the protein are in disordered regions?
- Compositional bias: what % of amino acids of the protein are compositionally biased?
- Sequence length: binned into percentiles, scaled from 0 to 1.³²

The data is drawn from Perdiggão et al. (2015), who constructed these measures using the 2014 release of Swiss-Prot.³³ To account for discrepancies with the 2020 release used in this study, I updated the measures following the same procedures as Perdiggão et al. (2015), where needed.

Below are features related to the importance of a protein:

³¹ Protein discovery year is the earliest recorded year of its existence, based on either its initial entry in major sequence databases (such as UniProt, the European Molecular Biology Laboratory database, and the DNA Data Bank of Japan) or its earliest mention in a publication, as documented by UniProt.

³² Longer proteins are generally more difficult to solve structurally, as they often exhibit increased complexity in expression and crystallization (Büssow et al. 2005). However, some very short proteins (e.g., those under 100 amino acids) can also pose challenges for protein production (Perdiggão et al. 2015; Slabinski et al. 2007). Since such sequences are rare in my dataset (less than 5%), I include sequence length as a positive component of the protein difficulty score (i.e., longer = more difficult). In my predicted brightness analysis—where I use Lasso Logit to predict whether a protein has a structure—multiple decile indicators of sequence length were selected, with the shortest deciles associated with a higher likelihood of being solved and the longest with a lower likelihood. This pattern supports the inclusion of sequence length as a proxy for difficulty.

³³ Available here: <https://www.pnas.org/doi/10.1073/pnas.1508380112#supplementary-materials>

- Number of publications about the protein: binned into percentiles, scaled from 0 to 1
- Is the protein targeted by a drug (binary)?³⁴
- Is the function of the protein known (binary)?
- Disease relevance (binary)
- Is the protein from human (binary)?

Data on importance was parsed from UniProt, except for drug information which was collected from DrugBank. While these importance features are conceptually relevant proxies, observed importance captures a mix of both the protein’s intrinsic scientific value and accumulated research effort. For example, a protein may play an important biological role that the scientific community has not yet discovered, which would not be captured by these proxies.

Since all of the features above are either naturally bounded or scaled to range from 0 to 1, I construct composite measures of difficulty and importance by taking the average of the respective features. This approach implicitly assumes equal weights and linearity across features—assumptions that may not fully capture the true relationship between these characteristics and the underlying constructs. One could, in principle, regress these features on whether a protein is bright or dark (as in the predicted brightness measure in Appendix A.6), but this conflates whether brightness reflects scientific importance or technical feasibility—which the regression cannot separately identify. Moreover, difficulty ideally would be measured by understanding how a protein’s physicochemical properties impact structure attempts and failures, which requires data on structure attempts. For this robustness analysis, however, the composite scores offer a transparent and interpretable approximation of technical difficulty and biological importance.

To aggregate to the cluster level, I take the average and standard deviation of the proteins’ difficulty and importance scores for each cluster. The resulting distributions are shown in Appendix Figure 2. Panel A displays the distribution of mean difficulty and importance across the 6,942 clusters. As in Table 1, it shows that bright clusters tend to include proteins that are both easier and more important. Panel B shows the distribution of the within-cluster standard deviation. Clusters show relatively little variation in difficulty (average SD = 0.01) but greater variation in importance (average SD = 0.17). This pattern follows from how clusters were constructed. As described in Section 4.3, clusters were constructed using sequence similarity

³⁴ Information on drugs is provided by DrugBank. This dataset provides comprehensive information on drugs at various development phases and their targets (i.e., proteins) and is freely available for academic use. A limitation of the free version of the data is that it only provides marketing dates for approved drugs, and there are no dates on when a drug entered pre-clinical or clinical trial phases.

because MR operates across proteins with similar amino acid sequences. Since sequence determines a protein’s physicochemical properties, they are correlated with technical feasibility. As a result, proteins within a cluster are likely to share difficulty but may still vary in importance.^{35,36} This is useful for the analysis in Appendix Table 4, where I split the clusters into “easy” clusters (those below the median in their composite difficulty score) and “hard” clusters (above the median), holding technical difficulty more constant while importance varies.

A.6 Predicting Brightness

The identification underpinning the difference-in-differences framework hinges on the parallel trends assumption. While there was no evidence of pre-trends in the event studies as well as in other robustness analyses, there may still be concerns over whether bright and dark clusters were evolving on different trends for factors unrelated to the introduction of MR. Proteins in bright clusters may be inherently more important or easier to solve, which could affect their trajectories, regardless of MR.

To address this concern, I develop a predicted brightness measure, where I measure whether a protein was predicted to be bright in 1998 (i.e. had a structure characterized),³⁷ using ex-ante traits related to both technical feasibility and biological importance.

Constructing Predicted Brightness

I restrict the sample to proteins that were discovered by 1998 ($n = 42,547$ proteins, of which 3% had a structure characterized by 1998). For each protein, I compiled features related to both technical feasibility and biological importance as of 1998, using more granular versions of those summarized in Table 1.

Specifically, technical feasibility measures include: (i) % of amino acids in membrane regions, (ii) % of amino acids in disordered regions, (iii) % of amino acids with compositional bias, and (iv) sequence length, binned into deciles.

³⁵ While sequence similarity can correlate with both structural and functional features, it more directly reflects a protein’s physicochemical properties, which influence technical feasibility. In contrast, biological importance is shaped by broader functional context and can vary substantially even among proteins with similar sequences (Hegyi and Gerstein 1999; Pearson 2013). As a result, clusters formed based on sequence similarity are likely to be more homogeneous in difficulty, but more heterogeneous in importance.

³⁶ Part of the higher within-cluster variation in importance may also reflect measurement limitations. While the importance features are conceptually meaningful, they likely contain more measurement error than the difficulty features, since observed importance partly reflects accumulated research attention in addition to intrinsic scientific value. This measurement noise could inflate within-cluster variance.

³⁷ Recall that (actually) bright clusters are defined as whether they had a structure by 1998.

Biological importance measures include: (i) the number of publications written about the protein (binned into 8 indicators), (ii) the number of drugs targeting the protein (binned into 6 indicators), (iii) whether the protein’s function was known (binary), (iv) whether the protein has disease relevance (binary), and (v) species (85 indicators).³⁸

I also control for discovery year using 29 indicators (with a single category absorbing all pre-1970 years). All continuous variables were standardized, and binning thresholds were selected to avoid extreme sparsity when creating indicator variables from categorical data. In total, the model includes 143 features.

To estimate the likelihood that a protein was structurally characterized by 1998, I fit a Lasso-penalized logistic regression model. The model was trained with 5-fold cross-validation on the full sample, selecting the penalty level that minimized average log loss across folds. After identifying the optimal penalty, the final model was fitted on the entire dataset. Of the 143 input features, the model selected 28 with non-zero coefficients.

As shown in Appendix Figure 3A, the model performance was evaluated using both cross-validated (out-of-sample) predictions and in-sample predictions from the final model. The cross-validated Receiver Operating Characteristic Area Under Curve (ROC AUC)³⁹ is 0.91, indicating strong ability to distinguish between bright and dark proteins. Given that bright proteins were rare as of 1998, the Precision-Recall AUC (PR AUC) may provide a more appropriate measure of performance, as it focuses on the model’s ability to correctly identify rare positive cases. The model achieves a PR AUC of 0.24, eight times larger than the base rate of 0.03, which is what would be expected under random guessing. The log loss, which evaluates how well-calibrated the predicted probabilities are by penalizing overconfident or incorrect predictions, is 0.09—30% improvement over the log loss of 0.13 that would be obtained under the base rate.⁴⁰ In-sample performance is slightly stronger, but the small gap between in-sample and out-of-sample metrics indicates that the model is not overfitting.

Appendix Figure 3B shows the distribution of this predicted brightness, by whether the protein was actually bright by 1998. The variation and overlap in predicted brightness between actually bright and dark proteins is useful: it allows for comparisons between clusters that were

³⁸ Species was defined at the genus level. If a genus had fewer than 100 observations, the phylum level was used instead. Species with fewer than 10 observations even at the phylum level were coded as “other.”

³⁹ ROC AUC can be interpreted as the probability that a random actually bright protein will have a higher predicted brightness than a random actually dark protein. The AUC can range from 0 to 1.

⁴⁰ Random guessing log loss can be calculated as $-(p \cdot \ln(p) + (1-p) \cdot \ln(1-p))$, where p is the proportion of positive cases (bright proteins). Lower log loss is better.

similar ex ante but differed in realized brightness. This makes predicted brightness a useful control, isolating the effect of MR from underlying differences in observable protein characteristics.

Finally, from this protein-level prediction, I aggregate up to the cluster-level by taking the average of the predicted brightness of all proteins in each cluster. I then use this cluster-level predicted brightness in two approaches: (i) including it as a control in the difference-in-differences specification and (ii) using it for matching.

Inclusion of Predicted Brightness in the Difference-in-Differences

I modify my baseline difference-in-differences framework to additionally control for predicted brightness. Specifically, I adapt Equation 1 to estimate the following:

$$Y_{ct} = \beta_0 + \beta_1 PostMR_t \times Bright_c + \beta_2 PostMR_t \times Predicted_Bright_c + \delta_t + \gamma_c + \varepsilon_{ct} \quad (2)$$

Equation 2 compares clusters of proteins that are similarly predicted to have their structures characterized by 1998 because they are ex-ante similar in traits related to biological importance and technical feasibility, but differ in whether they were actually structurally characterized.

The inclusion of predicted brightness helps isolate the added effect of actual brightness. If only predicted brightness is associated with post-MR solving activity (i.e., β_1 is non-significant but β_2 is significant), this would suggest that underlying characteristics are driving bright clusters to both have had their structures characterized in 1998 and subsequent structure characterization after MR. However, if there is an added effect of being actually bright in addition to being predicted bright (i.e., β_1 is significant), then this reduces the concern of omitted variable bias.

Appendix Table 6A reports the results from estimating Equation 2. The coefficients on $PostMR_t \times Bright_c$ remain positive and significant; among clusters predicted to be similarly bright, there is still an effect of being actually bright. With the exception of Column 3, the coefficient on $Post-MR \times Predicted_Bright_c$ is insignificant. This pattern reinforces the idea that while predicted brightness captures observed ex-ante traits, it cannot substitute for the presence of actual structural data, which MR requires to operate. These results support the interpretation that MR played a causal role in increasing structure solving in actually bright clusters, above what would be expected based on ex-ante observables.

Propensity Score Matching on Predicted Brightness

As an additional robustness check, I leveraged propensity score matching to construct a more comparable control group. I implemented a one-to-one nearest-neighbor matching on the

predicted brightness score, without replacement. Unmatched dark clusters were excluded. The resulting matched sample includes all bright clusters and a subset of dark clusters with similar predicted brightness. I then re-estimated Equation 1 using this smaller, matched sample. As reported in Appendix Table 6B, results remain consistent with those from the full sample, suggesting that the findings are not driven by the imbalance in ex-ante observables.

A.7 Alternative Sample Construction

Unbalanced panel: I restrict the sample to include all clusters that were discovered by 2003 (the year MR was introduced), rather than 1998 (the year before my sample begins) as in the main specification. This yields a larger, unbalanced panel of 12,292 clusters. As shown in Appendix Table 7A, results remain similar.

Similarity families: I explore an alternative classification of protein groupings by using UniProt’s curated protein families.⁴¹ Families are defined using sequence similarity as well as structural and functional similarities when available. Unlike the main clustering approach, which is based on an algorithmic threshold for sequence similarity defined by the user, the UniProt family classification provides a more “stable” partition of the protein universe since they are predefined and curated by biologists. These families often capture deeper evolutionary and functional relationships; however, the disadvantage is that they may exhibit lower sequence similarity. Following the same inclusion criteria as in my main specification, I restrict the sample to UniProt protein families that existed as of 1998 and contain at least one human protein.⁴² As shown in Appendix Table 7B, results remain similar.

A.8 Identifying Principal Investigators

Assigning Author IDs

Identifying the principal investigator (PI) behind each structure is a challenging task. The PDB does not provide author identifiers and only reports the authors’ last names and the initial

⁴¹ The list of families can be accessed here: https://www.uniprot.org/help/family_membership.

⁴² Some proteins are not assigned to any UniProt family. For these cases, I explore two approaches: treating them as singleton families or dropping them from the sample. In both cases, the results remain similar. Appendix Table 7B reports the version that includes the singleton families. Additionally, approximately 1% of proteins are classified into multiple families. For these proteins, I randomly assign each to a single family to maintain mutually exclusive units of analysis.

of first names, which makes it difficult to distinguish between PIs, especially those with common last names. To address this, I relied on Torvik and Smalheiser (2021)’s Author-ity data, which provides disambiguated author IDs for all articles indexed in PubMed, the leading database of life science articles.⁴³ This allowed me to identify the author IDs for all PDB structures that had a publication. Following scientific norms, I defined PIs as those who appear as the last author.

22% of structures, however, do not have publications and therefore cannot be directly linked to the Author-ity data. For these unpublished structures, I relied on a combination of automated matching and extensive manual cleaning to match an author ID to each structure. Specifically:

1. I first grouped all structures by the last authors’ last names and first name initials.
2. If the last name and the first name initial were associated with a single author ID in the Author-ity data among structures with publications, I applied the author ID to the other unpublished structures that share the same last name and first name initial.
 - i. It is possible that a PI may deposit a few structures in the PDB but never write any PDB paper. In this case, I would mistakenly view two PIs as the same PI. To reduce this likelihood, I manually reviewed PIs whose last names are common and ensured that they appear to be the same PI by reviewing co-authors, structure titles, organisms they study, middle initial if available, and years of activity in their publication history. If it appeared that the unpublished structure’s PI is in fact a different scientist, I returned to the entirety of the Author-ity data and looked up all authors that share the same last name and first name initial and manually identified the most likely match.
3. For last name and first name initial that were associated with multiple author IDs among published structures: I manually reviewed the unpublished structures to assign the appropriate author ID using similar procedures as 2i.
4. For PIs whose last name and first name initials appear on structures but not on any published PDB papers: these are likely PIs who peripherally engaged in structural biology by depositing a few structures in the PDB, but their research focus was outside of structural biology. To find author IDs for them, I followed the procedures below:
 - i. I pulled all potential matches that share the same last name and first name initial in the Author-ity data.

⁴³ Due to data availability of Torvik and Smalheiser (2021), the sample period for my analyses on PI runs from 1999-2016, instead of 2019.

- ii. I searched if any of these potential matches had the first author of the structure as a co-author to narrow the potential matches. I then identified the most likely match by reviewing years of publication activity and publication keywords (MeSH) that involves proteins.
- iii. For those with common last names with too many potential matches, no author ID was assigned to limit false matches.

Using these procedures, I identified author IDs for 98% of structures.

Identifying Affiliations

To identify affiliation, I leveraged Torvik (2021)’s MapAffil data, which standardized free-text affiliations available in PubMed articles.⁴⁴ PubMed started providing affiliation data of first authors in 1988 and affiliation for all authors in 2014. As a result, affiliation coverage is limited before 1988 (MapAffil merged on complementary sources, where available), and only the affiliation data of first authors is available before 2014. I followed the procedures below to identify affiliations:

1. For structures with publications, I used the affiliation of the last author, where available, and the first author if not. Because the norm in science is such that the first author is usually the lead graduate student or the post-doc, it is reasonable to assume that the first and last authors would share the same affiliation.
2. For structures without publications: since I have the author ID, I identified other papers by the same author in the year to infer the author’s affiliation.

Using these procedures, I identified affiliations for 87% of structures.

⁴⁴ The benefit of using the MapAffil data is that it provides Global Research Identifier Database (GRID) codes. This allows me to obtain standardized affiliations from which to match to the 2023 QS World University Rankings in Life Sciences and Medicine to identify PIs from elite vs. non-elite institutions. One limitation of relying on this ranking is that it does not rank elite research institutions that are not universities.

Appendix B. Solving the Phase Problem

Elucidating a protein structure requires crystallizing the protein and exposing the crystal to X-rays. This generates diffraction patterns as X-rays reflect off the crystal. Using a combination of statistical and physical principles, scientists analyze the diffraction data to construct a 3D model of the protein structure.

The “phase problem” is a critical challenge in this process. While X-ray reflections contain both amplitudes and phases, only the amplitudes can be directly measured from diffraction patterns. Without the phase information, the protein structure cannot be reconstructed.

This appendix details three methods used to solve this phase problem.⁴⁵ Two experimental phasing methods—*isomorphous replacement* and *anomalous diffraction*—solve the phase problem *de novo*, from scratch. These methods do not require templates of previously solved structures but are complex and labor-intensive. The third method, *molecular replacement*, is a computational approach that bypasses experimental phasing by leveraging existing templates.

B.1 Isomorphous Replacement

The oldest experimental method to solve the phase problem *de novo* is *isomorphous replacement*. This method involves making a known change to the target protein and analyzing how the change affects diffraction patterns. Structural biologists must produce at least two different types of crystals: a “native” target crystal and a “derivative” crystal with a heavy metal ion introduced. By comparing the diffraction patterns between the native and the derivative crystals, structural biologists can locate the heavy atoms and deduce possible phases of the other atoms in the proteins. Isomorphous replacement typically necessitates multiple derivative crystals with different heavy metals to fully solve the phase problem.

Isomorphous replacement can be laborious. Crystallizing proteins is a challenging process, and isomorphous replacement requires multiple derivations of the crystal, increasing the challenge involved. Furthermore, the metal ion must be introduced in a way that does not disturb the protein structure (i.e., maintain isomorphism), which can be difficult to achieve (Foos, Rizk, and Nanao 2022) and necessitates many rounds of trial-and-error.

⁴⁵ To construct this appendix, I consulted and adapted various sources. In particular, Cowtan (2003), Read (2005), and Terwilliger et al. (2016) were especially helpful.

B.2 Anomalous Dispersion

The second experimental method to solve the phase problem *de novo* is anomalous dispersion, where structural biologists vary the X-ray wavelength to induce atoms of specific elements in the protein to produce anomalous scattering. While the scattering of atoms is usually independent of the wavelength of the X-ray, at their respective “absorption edges,” each atomic type produces anomalous scattering, which introduces differences in the intensities of certain diffraction spots. By identifying the positions of these anomalous scattering atoms, structural biologists gain clues to recover the missing phase information of the rest of the protein.

While anomalous dispersion has advantages over isomorphous replacement (in particular, anomalous dispersion only requires a single crystal), it comes with its own challenges. The experiment has to be conducted at high precision because signals from anomalous dispersion are weaker than those from isomorphous replacement (Leonard 2018). Because small differences in diffraction patterns must be detected, anomalous dispersion experiments require access to synchrotron facilities that can provide tunable X-ray beams at high intensities (Oksanen and Goldman 2010). Moreover, radiation damage to the crystal is a concern when structural biologists need to tune the X-ray wavelength multiple times, as many sets of data are collected from a single crystal.

B.3 Molecular Replacement

Molecular replacement (MR) allows structural biologists to bypass the time-intensive and challenging experimental phasing methods discussed above, eliminating the need for additional experimental steps. Rather than solving the phase problem *de novo*, MR imports phase information from existing structures that are similar to the unknown target structure.

MR proceeds in two steps: (i) identify previously solved structures that can be used as templates and (ii) orient the template structure to match the position of the unknown target structure within the crystal. After these two steps, the phase information can be transferred from the template to the target protein.

First, the key insight behind molecular replacement rests on an empirical pattern. Proteins are composed of sequences of amino acids, and sequence similarity has been observed to be highly correlated with structural similarity. Although the exact mechanisms of how specific amino acid sequences determine a protein’s 3D shape are not fully understood, this correlation allows structural biologists to leverage MR.

Specifically, because sequence similarity is correlated with structure similarity, structural biologists can reasonably assess whether a previously solved structure will serve as a suitable template for the unknown target protein based on sequence similarities. The closer the sequence similarity, the more likely that the previously solved structure and the unknown structure of the target protein are also analogous. As a general rule of thumb, MR will likely work if the template and target proteins share at least 30% sequence identity. If the sequence identity falls below 30%, MR will be usually challenging to implement (Schmidberger et al. 2010; Phenix 2022).⁴⁶

Second, to import the phase information from the known template to the unknown target protein, the template protein must be correctly oriented and positioned within the repeating unit of the target protein crystal. The template structure is first rotated in three dimensions, and the resulting amplitudes and phases are calculated for every orientation. The software program Phaser uses maximum likelihood to identify the orientation that best matches the observed experimental diffraction patterns of the unknown target protein. The oriented template structure is then placed at every possible position in the unit cell. Again, the position that best matches the observed experimental diffraction patterns is chosen.

At the end, Phaser outputs a list of possible solutions with a signal-to-noise metric called the “Z-score,” which provides guidance on whether the solution has been identified. Once the correct orientation and position of the template structure are found, phase information can be directly imported into the unknown target protein.

Molecular Replacement vs. Machine Learning

While MR and machine learning methods are both data-extrapolation approaches that rely on the existence of prior data from which they import information, the principles behind MR and machine learning are distinct: MR is based on a pre-specified model grounded in the laws of physics, whereas machine learning learns the model directly from the data without such pre-specified models.

Machine learning takes training data as input, infers a model from the statistical patterns in the training data, and then applies the model in unseen test data. In contrast, MR does not “learn” a model from the data like machine learning; it instead relies on the physics of X-ray diffractions to guide the process. MR involves solving equations that describe the interaction of

⁴⁶ MR success is not always guaranteed even with high sequence similarity due to factors such as data quality and conformational changes.

X-rays with the protein crystal lattice. Using numerical optimization, MR determines the correct orientation and position of a template protein within the unit cell to match the target protein.

While the specifics of *implementing* MR come from models of physics, the ability to *use* MR stems from empirical patterns. Although the exact mechanisms of how amino acid sequences determine 3D structures remain unclear, structural biologists leverage the correlation between sequence and structural similarity. When structural biologists select which structure to solve, they can reasonably assess whether MR is feasible based on the availability and sequence of previously solved structures. Above all, like machine learning, the key limitation of MR is that it can only work by analogy to known structures in the PDB and cannot be applied without prior solved templates of sufficient similarity.